

# Breast Cancer Follow-Up and Surveillance After Primary Treatment: ASCO Guideline Update

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## ABSTRACT

ASCO Guidelines provide recommendations with comprehensive review and analyses of the relevant literature for each recommendation, following the guideline development process as outlined in the <https://www.asco.org/practice-patients/guidelines/guideline-methodology>. ASCO Guidelines follow the <https://www.asco.org/about-asco/legal/conflict-interest-coi>.

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**PURPOSE** To offer clinicians recommendations concerning breast cancer follow-up and surveillance after primary treatment.

**METHODS** ASCO convened an Expert Panel to develop recommendations based on a systematic review and a formal consensus process.

**RESULTS** One randomized controlled trial (RCT) was identified and formed the evidentiary basis for the surveillance mammography guideline recommendation. No RCTs were identified that evaluated different follow-up approaches by risk of relapse in patients treated for early-stage breast cancer. Given the dearth of evidence identified in the systematic review of the literature, formal modified Delphi consensus-based recommendations were generated.

**RECOMMENDATIONS** The guideline offers recommendations on the role of a risk-based approach to the frequency and intensity of follow-up among patients with breast cancer in the adjuvant setting. Regular history, physical examination, and mammography are recommended for breast cancer follow-up. Either virtual or in-person visits can be offered. Certain patients at a high risk of cancer recurrence warrant closer surveillance, such as those patients with locally advanced disease or residual disease following neoadjuvant chemotherapy. Recommendations on the use of blood-based biomarkers and supplemental imaging are provided. Guideline recommendations will be updated once additional evidence-based tools become available to better individualize surveillance.

Additional information is available at [www.ascopubs.org/topics/asco-guidelines/breast-cancer](http://www.ascopubs.org/topics/asco-guidelines/breast-cancer).

## ACCOMPANYING CONTENT

 Appendix  
 Data Supplement

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## INTRODUCTION

The purpose of this guideline is to offer clinicians recommendations on breast cancer post-treatment surveillance.

ASCO first published an evidence-based clinical practice guideline on breast cancer follow-up and management in asymptomatic patients after primary, curative therapy in 1997.<sup>1</sup> That guideline was updated in 1999<sup>2</sup> and again in

2006<sup>3</sup> and 2013.<sup>4</sup> This update offers recommended follow-up care in the context of current clinical practice and provides clinicians and other health care practitioners, patients, and caregivers with formal consensus-based recommendations regarding risk-based surveillance management for patients who have been previously treated for early-stage breast cancer.

## **GUIDING CONCEPTUAL FRAMEWORK: RISK-BASED POST-TREATMENT SURVEILLANCE**

In this guideline, the Expert Panel adopted a risk-stratified approach to post-treatment surveillance for breast cancer. By risk, the Panel is referring to local, regional, and/or distant recurrence. (Emerging studies should allow more precise prediction of recurrence risk, but it is premature to include such estimates in the present guideline.) These approaches are increasingly being considered in the settings of cancer prevention, treatment, and post-treatment survivorship. The goal of risk-stratified approaches is to address better the heterogeneity of patient needs in ways that account for patient preferences, clinical workforce shortages, and rising health care costs.<sup>5,6</sup> The United States is facing a critical shortage in the oncology workforce relative to the number of patients who will need cancer care. ASCO documented the projected oncologist shortages in 2007<sup>7</sup> and 2014<sup>8</sup>; these have been exacerbated by the COVID-19 pandemic, oncologist retirements, and the rising numbers of survivors. The shortage is particularly impactful for rural America, where projections suggest that oncologists will be able to meet the needs of only 29% of patients by 2037.<sup>9,10</sup>

Risk-stratified approaches to care that vary the intensity or frequency of care, clinical setting, and expertise of the clinician by patient risk or need, have been proposed as one way to meet the needs of patients while most efficiently using resources. For post-treatment follow-up, specifically, there have been calls by ASCO, the American Cancer Society (ACS), and others to consider ways to decrease the intensity of care for lower-risk survivors, or to transition them from oncology to survivorship teams or primary care, where possible, to free up oncology visits for patients with active cancer or those with higher risk or acuity.<sup>5,11</sup> Such risk-stratified approaches to cancer follow-up care have been implemented in England, Northern Ireland, and Australia<sup>11</sup>; evaluation of this approach in the United Kingdom found it met patients' needs while freeing up oncologist slots, reducing hospital visits, and decreasing overall health care costs.<sup>12,13</sup> Based on this success, the United Kingdom adopted a risk-stratified approach to their national survivorship initiative, and there have been calls to test and evaluate similar approaches in the United States and across the globe.<sup>6,11,14</sup>

It is ideal, of course, for proposed clinical practice changes to be informed by evidence from high-quality randomized controlled trials (RCTs); however, such data do not yet exist. Although studies of risk-stratified or alternative approaches to breast cancer follow-up are ongoing (Table 1), there is a

pressing and immediate need to offer guidance to most efficiently care for today's patients.

## **GUIDELINE QUESTIONS**

This clinical practice guideline update addresses one overarching clinical question: What is the role of a risk-based approach to the frequency and intensity (type of and number of tests) of follow-up among patients with varying risks of breast cancer recurrence in the adjuvant setting after curative-intent primary therapy? Within that broad question, the guideline provides recommendations on (a) intensity-based post-treatment surveillance, (b) the use of blood-based biomarkers in breast cancer post-treatment surveillance, and (c) supplemental surveillance imaging.

## **METHODS**

### **Guideline Development Process**

This systematic review-based, formal consensus guideline was developed by a multidisciplinary Expert Panel, which included a patient representative, a representative from ACS, and an ASCO guidelines staff member with health research methodology expertise (Appendix Table A1, online only). The recommendations were based on clinical experience and on limited evidence published after the previous guideline, identified through online searches of PubMed (January 1, 2015, to March 2, 2026). A series of searches was conducted to inform the guideline recommendations, one to identify reports of RCTs and meta-analyses of RCTs that evaluated different follow-up approaches by risk of relapse and two targeted searches on, respectively, surveillance imaging in this patient population and the clinical utility of circulating tumor DNA (ctDNA) testing for recurrence monitoring (see Data Supplements 1A through 2C for more details of the searches). ASCO's Circulating Tumor DNA Testing in Solid Tumors and Lymphoma guideline recommendations<sup>15</sup> were adapted for the present update and the supporting literature review and analysis from that guideline relevant to ctDNA in monitoring recurrence are referenced here.

Articles were selected for inclusion in the systematic review based on the following criteria:

- Population: patients with breast cancer who have completed primary therapy with curative intent who are without clinical evidence of disease
- Interventions: modes of surveillance for patients with breast cancer who have completed primary therapy with curative intent
- Comparisons: alternate modes of surveillance (eg, more or less intensive) for patients with breast cancer who have completed primary therapy with curative intent by risk of recurrence
- Outcomes: disease-free survival, overall survival, health-related quality of life (QoL), treatment toxicity, and anxiety related to testing and fear of recurrence

**TABLE 1. Ongoing Trials**

| Trial Registry Number                         | Patient Population                                                                | Inclusion Criteria                                                                                                                                                                                                                                                                                                                                                                        | Planned Accrual | Accrual Start | Primary End Point                                                                                 | Intervention                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Comparator                                                                                                                                                                                        | Current Status         |
|-----------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| INSPIRE<br>JCOG1204<br>UMIN000012429          | Patients with high-risk breast cancer (expected 5-year recurrence rate of >30%)   | For patients who did not receive preoperative chemotherapy: ≥10% ER staining by IHC and ≥4 axillary nodal metastases OR <10% ER staining and ≥1 axillary nodal metastases<br>For patients who received preoperative chemotherapy: ≥10% ER staining by IHC and ≥4 axillary nodal metastases OR <10% ER staining and histologically proven residual invasive cancer in the breast or axilla | 1,700           | Nov 2013      | OS                                                                                                | Intensive follow-up<br>Mammography once a year<br>Physical examination every 3 months for 3 years, then every 6 months for 2 years, followed by once a year<br>Tumor marker evaluation every 3 months for 3 years, then every 6 months for 2 years<br>Chest and abdominal CT, bone scintigraphy, <sup>a</sup> and brain MRI twice a year for 3 years, then once a year for 2 years                                                                                                                                                                                   | Standard follow-up<br>Mammography once a year<br>Physical examination every 3 months for 3 years, then every 6 months for 2 years, followed by once a year<br>Tumor marker evaluation once a year | Accrual complete       |
| NABOR<br><a href="#">NCT05975437</a>          | Patients who received curative treatment for non-metastatic primary breast cancer | New, female patients, age ≥40, curatively treated for invasive, non-metastasized primary breast cancer, starting follow-up care                                                                                                                                                                                                                                                           | 1,040           | Mar 2023      | Patients' experienced level of cancer worry and overall self-rated quality of life (EQ-VAS-score) | Personalized Surveillance Plan (PSP)<br>Generated by the Breast Cancer Surveillance Decision Aid, incorporating the INFLUENCE prediction tool 3.0<br>Estimated personal risk generated and patient completes decision aid<br>Summary of decision aid and results of first surveillance mammogram used for shared decision-making<br>Personalized After Care Plan (PAP)<br>Patients complete aftercare decision aid, including needs assessment<br>Summary of decision aid used in shared decision making<br>PAP can be re-evaluated and changed throughout aftercare | Care as usual (CAU)<br>For the majority of hospitals<br>Mammography and physical examination annually<br>Discussion of results and aftercare consultations                                        | Enrolling participants |
| REaCT-Wellness<br><a href="#">NCT05365230</a> | Patients who have completed acute phase of treatment for invasive breast cancer   | Eligible participants were patients with stage I-III breast cancer who had completed the acute phase of their breast cancer treatment (ie, surgery, chemotherapy/HER2-based therapy, and radiotherapy) and began endocrine therapy (if applicable)                                                                                                                                        | 228             | Aug 2022      | HRQOL, assessed with FACT-G                                                                       | On-demand personalized follow-up care<br>Mammography once a year<br>On-demand access to a Wellness Beyond Cancer Program nurse and annual follow-up by phone following annual mammogram                                                                                                                                                                                                                                                                                                                                                                              | Guideline-based follow-up care<br>Mammography once a year<br>Follow-up care based on current guidelines (standard of care)                                                                        | Accrual complete       |

Abbreviations: CAU, care as usual; CT, computed tomography; EQ-VAS, EuroQol Visual Analogue Scale; ER, estrogen receptor; FACT-G, Functional Assessment of Cancer Therapy—General; HER2, human epidermal growth factor receptor 2; HRQOL, health-related quality of life; IHC, immunohistochemistry; LHRH, luteinizing hormone-releasing hormone; MRI, magnetic resonance imaging; OS, overall survival; PAP, personalized aftercare plan; PSP, personalized surveillance plan.

## TARGET POPULATION AND AUDIENCE

### Target Population

Patients with breast cancer who have completed primary therapy with curative intent and who are without clinical evidence of disease.

### Target Audience

Oncology specialists, other health care clinicians (including primary care physicians, advanced practice providers, specialists, survivorship care professionals, nurses, social workers, and any other relevant member of a comprehensive multidisciplinary cancer care team), caregivers, and patients.

Articles were excluded from this systematic review if they were (1) meeting abstracts not subsequently published in peer-reviewed journals; (2) editorials, commentaries, letters, news articles, case reports, and narrative reviews; and (3) published in a non-English language.

Because of the limited high-quality evidence available to inform the clinical questions, recommendations were developed using the ASCO modified Delphi formal consensus methodology.<sup>16</sup> This process involved the drafting of recommendations by an Expert Panel (N = 16). These preliminary recommendations represent the best clinical opinion of the Expert Panel based on personal experience in the management of breast cancer. The Expert Panel met by conference call to review and refine the recommendations. The Expert Panel was then supplemented by additional experts (N = 17), who, with the Expert Panel members, were recruited to rate their agreement with the recommendations. The entire membership of 33 experts is referred to as the Consensus Panel (Appendix Table A2). Each recommendation had to be agreed to by at least 75% of Consensus Panel respondents to be accepted. This methodology is described in further detail elsewhere (<https://www.asco.org/practice-patients/guidelines/guideline-methodology>).

Two full panel meetings were held, and members were asked to provide ongoing input on the updated guideline development protocol, quality and assessment of the evidence, generation of recommendations, draft content, and review and approve drafts during the entire development of the guideline. ASCO staff met routinely with the Expert Panel co-chairs and corresponded with the Panel via e-mail to coordinate the process to completion. Ratings for the strength of the recommendation and evidence quality are provided with each recommendation, defined in Appendix Table A3. The quality of the evidence for each outcome was assessed using the Cochrane Risk of Bias tool and elements of the GRADE quality assessment and recommendations

development process.<sup>17,18</sup> GRADE quality assessment labels (ie, high, moderate, low, very low) were assigned for each outcome by the project methodologist in collaboration with the Expert Panel co-chairs and reviewed by the full Expert Panel. All funding for the administration of the project was provided by ASCO.

## Guideline Review and Approval

The draft recommendations were released to the public for open comment from February 9, 2026 through February 23, 2026. Response categories of Agree as written, Agree with suggested modifications, and Disagree, see comments were captured for every proposed recommendation with 47 written comments received. A total of 63% of the 19 respondents either agreed or agreed with slight modifications to the recommendations, and 37% of the respondents disagreed with one or more of the recommendations. Expert Panel members reviewed comments from all sources and determined whether to maintain original draft recommendations, revise with minor language changes, or consider major recommendation revisions.

All changes were incorporated into the final manuscript before ASCO Evidence Based Medicine Committee (EBMC) review and approval. All ASCO guidelines are ultimately reviewed and approved by the Expert Panel and the ASCO EBMC before submission to the *Journal of Clinical Oncology* for editorial review and consideration for publication.

## Guideline Updating

The ASCO Expert Panel and guidelines staff will work with co-chairs to keep abreast of any substantive updates to the guideline. Based on formal review of the emerging literature, ASCO will determine the need to update. The ASCO Guidelines Methodology Manual (available at <https://www.asco.org/practice-patients/guidelines/guideline-methodology>) provides additional information about the guideline update process. This is the most recent information as of the publication date.

## RESULTS

### Characteristics of Studies Identified in the Literature Search

A total of 2,483 abstracts were identified in the broad literature search conducted to identify reports of RCTs and meta-analyses of RCTs that evaluated different follow-up approaches. After applying the eligibility criteria, one RCT remained, forming the evidentiary basis for the surveillance mammography guideline recommendation.<sup>19</sup> The identified trial was published in 2025. Characteristics of the trial participants are provided in Table 2; key findings are reported in Table 3. No RCTs were identified that evaluated different follow-up approaches by risk of relapse in patients treated for early-stage breast cancer.

**TABLE 2.** Characteristics of Patients and Disease in Mammo-50

| Study Country                  | Interventions and Comparisons                                                                                                                  | Patients, No. | Patient Characteristic |                                                                                                                                                               | Disease Characteristic                                                                                                                                                 |                                                  |                                                                                                                                                                                                          |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                |                                                                                                                                                |               | Median Age, Years      | Race or Ethnicity                                                                                                                                             | Type of Disease                                                                                                                                                        | Surgery Type                                     | Receptor Status                                                                                                                                                                                          |
| Mammo-50<br>The United Kingdom | Less frequent mammography (biennial mammography for patients with conservation surgery and triennial mammography for patients with mastectomy) | 2,617         | 66                     | White: 96.8%<br>Mixed: 0.5%<br>Asian or British: 1.0%<br>Asian: 1.0%<br>Black or Black British: 0.7%<br>Chinese or other ethnic origin: 0.1%<br>Unknown: 1.0% | DCIS only: 12.5%<br>Invasive (with or without DCIS): 87.5%<br>Axillary node negative: 72.3%<br>Histologic grade<br>1: 20.7%<br>2: 51.2%<br>3: 27.3%<br>Not known: 0.8% | Conservation surgery: 80.2%<br>Mastectomy: 19.8% | ER<br>Negative: 11.1%<br>Positive: 82.9%<br>Not evaluated: 6.0%<br>PR<br>Negative: 19.2%<br>Positive: 45.1%<br>Not evaluated: 35.7%<br>HER2<br>Negative: 86.9%<br>Positive: 11.8%<br>Not evaluated: 1.4% |
|                                | Annual mammography                                                                                                                             | 2618          | 66                     | White: 97.4%<br>Mixed: 0.5%<br>Asian or British: 0.6%<br>Asian: 0.6%<br>Black or Black British: 0.8%<br>Chinese or other ethnic origin: 0.2%<br>Unknown: 0.4% | DCIS only: 12.6%<br>Invasive (with or without DCIS): 87.4%<br>Axillary node negative: 75.1%<br>Histologic grade<br>1: 21.9%<br>2: 53.4%<br>3: 23.7%<br>Not known: 1.0% | Conservation surgery: 80.3%<br>Mastectomy: 19.7% | ER<br>Negative: 10.7%<br>Positive: 82.5%<br>Not evaluated: 6.8%<br>PR<br>Negative: 17.8%<br>Positive: 47.7%<br>Not evaluated: 34.5%<br>HER2<br>Negative: 87.5%<br>Positive: 11.1%<br>Not evaluated: 1.3% |

Abbreviations: DCIS, ductal carcinoma in situ; ER, estrogen receptor; HER2, human epidermal growth factor receptor 2; PR, progesterone receptor.

As mentioned, because of the limitations of the available evidence, the guideline relied on a formal consensus development process to generate practice recommendations. Respondents were asked to rate their agreement with the proposed definitions and features of the three intensity subgroups—low, intermediate, and high—and with recommendations for follow-up testing corresponding to these three subgroups. In addition, respondents rated their agreement with a recommendation concerning the use of CBC count, chemistry panels, and the tumor markers carcinoembryonic antigen (CEA), cancer antigen (CA) 15-3, CA 27.29 for routine surveillance of patients with breast cancer after completion of primary therapy and with a separate ctDNA recommendation that was adopted from the ASCO Circulating Tumor DNA Testing in Solid Tumors and Lymphoma guideline.<sup>15</sup> Finally, respondents rated their agreement with recommendations on use of post-treatment imaging surveillance for asymptomatic patients. The full Consensus Panel completed three rounds of voting.

During the first round, agreement with the individual recommendations ranged from 76% to 97% (N = 33). The number of respondents across recommendations ranged from 25 to 32 of the 33 Consensus Panel members. All the recommendations satisfied the required 75% agreement threshold. The Expert Panel Co-Chairs nonetheless chose to submit one of the round 1 recommendations for which there was a modest level of agreement (76%), to a second round of voting after revising the recommendation language based on feedback provided by Consensus Group members. Round 2

agreement with this revised recommendation was 82%. Even with this improved rate of agreement, reasonable concerns expressed about the implications of the recommendations for both clinicians and survivors occasioned a third round of voting after the Panel Co-Chairs further revised this recommendation to reflect the cogent comments and suggested revisions offered by several Consensus Group members. Round 3 agreement with this revised recommendation was 94%.

## RECOMMENDATIONS

All recommendations are available in [Table 4](#), and [Figures 1-4](#) show the recommendations as an algorithm.

## INTENSITY-BASED POST-TREATMENT SURVEILLANCE

This section addresses Recommendations 1.1.1 through 1.3.2.

### Literature Review

There is no direct evidence from RCTs to inform the clinical question regarding the role of a personalized or tailored approach to the frequency and intensity of follow-up among patients with varying risks of recurrence in the adjuvant setting after curative-intent primary therapy. The recommendations offered represent the best clinical opinion of the Expert Panel members based on personal experience in the management of breast cancer and the input offered by members of the Consensus Panel.

**TABLE 3. Mammo-50 Noninferiority Phase III RCT—Results Summary**

Population: Women age  $\geq 50$  years with breast cancer who were recurrence free 3 years after curative surgery  
 Intervention: Less frequent mammography (biennial mammography for patients with conservation surgery and triennial mammography for patients with mastectomy)  
 Comparator: Annual mammography

| Outcome Timeframe                          | Study Results and Measurements                                                                                            | Absolute Effect Estimates                                                                             |                                          | Quality of Evidence                                  | Summary                                                                                                                            |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|------------------------------------------|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
|                                            |                                                                                                                           | Annual Mammography                                                                                    | Less Frequent Mammography                |                                                      |                                                                                                                                    |
| Breast cancer–specific survival<br>5 years | HR, 0.92 (95% CI, 0.64 to 1.32)<br>Based on data from 5,235 participants in one study <sup>a</sup><br>Follow-up 5.7 years | 19 deaths due to breast cancer per 1,000<br>Difference: 2 fewer per 1,000 (95% CI, 7 fewer to 6 more) | 17 deaths due to breast cancer per 1,000 | Moderate<br>Due to serious risk of bias <sup>b</sup> | Less frequent mammography probably has little or no difference on breast cancer–specific survival compared with annual mammography |
| OS<br>5 years                              | HR, 1.07 (95% CI, 0.87 to 1.33)<br>Based on data from 5,235 participants in one study <sup>a</sup><br>Follow-up 5.7 years | 53 deaths per 1,000<br>Difference: 4 more per 1,000 (95% CI, 7 fewer to 17 more)                      | 57 deaths per 1,000                      | Moderate<br>Due to serious risk of bias <sup>b</sup> | Less frequent mammography probably has little or no difference on OS compared with annual mammography                              |
| 5-year recurrence-free interval<br>5 years | HR, 1.0 (95% CI, 0.81 to 1.23)<br>Based on data from 5,235 participants in one study <sup>a</sup><br>Follow-up 5.7 years  | 59 per 1,000<br>Difference: 0 fewer per 1,000 (95% CI, 11 fewer to 13 more)                           | 59 per 1,000                             | Moderate<br>Due to serious risk of bias <sup>b</sup> | Less frequent mammography probably has little or no difference on 5-year recurrence-free interval compared with annual mammography |

Abbreviations: HR, hazard ratio; OS, overall survival; RCT, randomized controlled trial.

<sup>a</sup>Primary study (1) baseline/comparator control arm of reference used for intervention.

<sup>b</sup>Risk of bias: serious. Only one, open-label study.

In the low-intensity surveillance group (Table 4, Section 1.1.1), the mammographic surveillance recommendations were informed in part by the results of the Mammo-50 noninferiority phase III RCT that evaluated annual versus less frequent mammography in women age  $\geq 50$  years after breast cancer diagnosis (Tables 2 and 3).<sup>19</sup> Dunn et al<sup>19</sup> randomly assigned 5,235 women recurrence free at 3 years after breast cancer diagnosis to either annual mammography (n = 2,618) or to less frequent mammography, biennial mammography for patients with conservation surgery, and triennial mammography for patients with mastectomy (n = 2,617). Cost-effectiveness and breast cancer–specific survival were the coprimary outcomes; number of referrals back to the hospital system, recurrence-free interval, and overall survival were the secondary outcomes.

At a median follow-up of 5.7 years, less frequent surveillance mammography was noninferior compared with yearly mammograms for recurrence-free interval, overall survival, and breast cancer–specific survival. Based on analyses of QoL outcomes and patient-reported outcomes, Marshall et al<sup>20</sup> reported that there were no differences between the annual v less frequent surveillance mammography arms of Mammo-50 in QoL scales and other measures of patient-reported experiences including distress level, worries and concerns, and fear of recurrence. As such, the Mammo-50 trial offers guidance for safely de-escalating mammography surveillance for select breast cancer survivors  $\geq 50$  years of age, as risk-stratified by the Expert Panel in the low-intensity group. For intermediate-intensity and high-

intensity surveillance groups recommendations refer to Table 4, sections 1.2.1 and 1.3.1 respectively.

### Clinical Interpretation

The optimal frequency and intensity of follow-up and management of patients with breast cancer who have completed primary therapy with curative intent remain a challenge for clinicians because of the absence of high-quality RCTs and the lack of validated prognostic models to tailor post-treatment surveillance. This is compounded by the well-documented worsening shortage in the US oncology workforce driven by many factors including an aging population and demand for services that is disproportionately rising by nearly 50%, while the supply of oncologists increased by around 15% in the past 5 years.<sup>7,9</sup>

Currently, there remain differences in the follow-up schedule, laboratory, and imaging investigations in the clinical practice setting, complicated further by clinicians ordering or patients requesting unnecessary tests not currently incorporated in any of the existing guidelines.<sup>21</sup> This risk-stratified guideline offers practical guidance to assist clinicians in efficiently caring for patients based on a multidisciplinary consensus process and addresses the current unmet need pending the availability of further published evidence. Biomarker testing, including liquid biopsy and ctDNA, to detect minimal residual disease (MRD) is an emerging area of interest for monitoring recurrence,

TABLE 4. Summary of All Recommendations

| Topic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Formal Consensus Recommendation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Agreement, % |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <p><i>General Notes.</i> The following recommendations (strong or conditional/weak) and terminology (see Data Supplement) represent reasonable options for patients depending on clinical circumstances and in the context of individual patient preferences. Recommended care should be accessible to patients whenever possible. The recommendations offered represent the best clinical opinion of the Expert Panel members based on personal experience in the management of female breast cancer. Follow-up care for patients who are treated for male breast cancer should generally be similar to care provided for patients with female breast cancer.<sup>99</sup> In addition, the ACR provides recommendations specifically for transgender persons assigned female at birth with a history of breast cancer.<sup>101</sup></p> <p><i>Note Concerning the Definitions and Features of the Proposed Intensity Subgroups.</i> When reviewing the following draft intensity-based recommendations for post-treatment surveillance, Consensus Group members were asked to rate agreement with both the proposed definition/features of the three intensity subgroups (low, intermediate, and high) and the proposed recommended follow-up testing.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |              |
| Intensity-based post-treatment surveillance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>1.1.1. Low-Intensity Surveillance Group</p> <p>Ductal carcinoma in situ (no endocrine therapy or off endocrine therapy)</p> <p>Stage I HR-positive breast cancer (off endocrine therapy and any other systemic therapy after completion of treatment)</p> <p>TNBC or HER2-positive breast cancer after at least 5 years from diagnosis (Evidence quality: NA; Strength of recommendation: Conditional)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 94           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>1.1.2. Annual clinical visit, in person or via virtual consultation, by survivorship clinics or nononcologists such as oncology advanced practice providers or within routine primary care or gynecology visits.</p> <p>Patients after breast-conserving surgery should be offered annual mammograms for the first 3 years after completion of treatment. Thereafter, surveillance mammography every 1-2 years is recommended for patients 50 years or older with low-risk features (T1, low-grade, node-negative, ER-positive tumors, who underwent BCS, and who are recurrence free at 3 years post diagnosis). The Expert Panel recommends annual surveillance mammography for patients &lt;50 years old or with TNBC or HER2-positive tumors.</p> <p>Continued annual surveillance mammography is an option for patients who would prefer more frequent mammography after discussion with their clinician regarding the potential harms (eg, increased anxiety due to false-positive findings, further investigations for additional findings) and potential benefits (eg, earlier detection of new or recurrent breast cancer, reassurance, reduced anxiety) of annual mammographic surveillance. (Evidence quality: Moderate; Strength of recommendation: Conditional)</p> <p><i>Qualifying Statement and Key Evidence for Recommendation 1.1.2:</i> These recommendations are based on expert consensus and on the findings of an RCT, Mamo-50<sup>19</sup> (median age = 66 years) that included patients who had predominantly had small, low-grade, ER-positive tumors, who underwent breast-conserving surgery, and who were recurrence free at 3 years post diagnosis. The effect of less-frequent screening did not vary significantly by breast cancer subtype, but the study was not powered to detect subgroup differences.</p> | 94           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>1.2.1. Intermediate-Intensity Surveillance Group</p> <p>Stage I TNBC or HER2-positive breast cancer within the first 5 years from diagnosis, stage II or stage III TNBC with pCR after completion of neoadjuvant therapy, stage II or stage III HER2-positive breast cancer with pCR after completion of neoadjuvant therapy</p> <p>Patients on ongoing endocrine therapy (any stage, including those who did not achieve pCR)</p> <p>Stage II or III ER-positive who did not take CDK 4/6 inhibitors or 5 years of endocrine therapy</p> <p>Patients with HR-positive breast cancer on endocrine therapy who completed targeted adjuvant CDK4/6 inhibitor therapy, then transition to low-Intensity group at the completion of endocrine therapy (Evidence quality: NA; Strength of recommendation: Conditional)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 87           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>1.2.2. Clinical follow-up (history and physical examination) with oncology team or cancer survivorship clinic every 6-12 months for 5-10 years after diagnosis. Following 5 years, patients can transition and be managed in the low-Intensity group by survivorship clinics with advanced practice providers or nononcologists, such as within routine primary care visits.</p> <p>Annual surveillance mammography is recommended for patients treated with breast-conserving surgery. For patients treated with unilateral mastectomy with or without reconstruction, annual contralateral surveillance mammography is indicated. For patients treated with bilateral mastectomy with or without reconstruction, mammography is not recommended unless warranted by clinical symptoms. For supplemental imaging, see Surveillance Imaging recommendations. (Evidence quality: Low; Strength of recommendation: Conditional)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 82           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>1.3.1. High-Intensity Surveillance Group</p> <p>TNBC with residual disease after neoadjuvant chemotherapy and within 5 years of diagnosis</p> <p>Patients with TNBC or HER2-positive breast cancer continuing on active systemic therapy</p> <p>Patients with germline mutations (eg, <i>BRCA1/2</i>, <i>PALB2</i>)</p> <p>HER2-positive breast cancer with residual disease after neoadjuvant chemotherapy and within 5 years of diagnosis</p> <p>Inflammatory breast cancer (any subtype) and within 10 years of diagnosis</p> <p>Patients with hormone receptor-positive breast cancer on endocrine therapy and on CDK4/6 inhibitor therapy (Evidence quality: NA; Strength of recommendation: Conditional)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 97           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>1.3.2. After completing active treatment (eg, targeted therapy, immunotherapy, other than endocrine therapy), clinical follow-up (history and physical examination) with the oncology team every 3-6 months for up to 10 years after diagnosis. Following that, patients can transition to the Low-Intensity group and be managed by survivorship clinics or nononcologists such as within routine primary care visits</p> <p>Annual surveillance mammography is recommended for patients treated with breast-conserving surgery. For patients treated with unilateral mastectomy with or without reconstruction, annual contralateral surveillance mammography is indicated. For patients treated with bilateral mastectomy with or without reconstruction, mammography is not recommended. Routine chest wall examinations are suggested every 6 months. For supplemental imaging, see Surveillance Imaging recommendations. (Evidence quality: Low; Strength of recommendation: Conditional)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 84           |

(continued on following page)

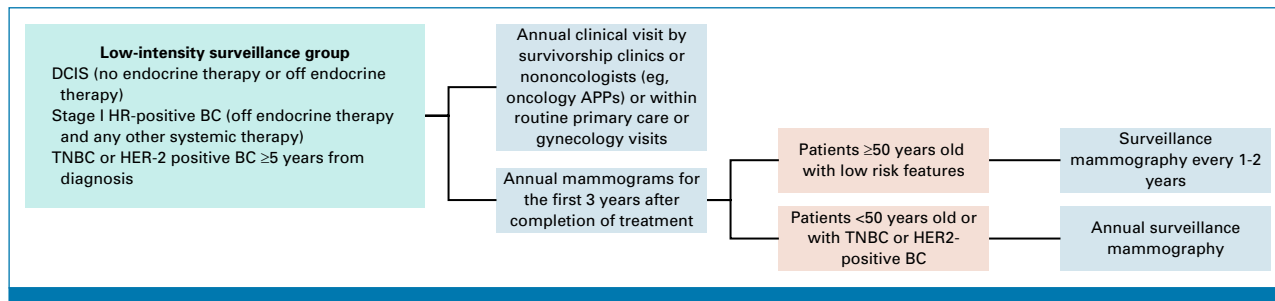
TABLE 4. Summary of All Recommendations (continued)

| Topic                                                                      | Formal Consensus Recommendation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Agreement, %                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use of blood-based biomarkers in breast cancer post-treatment surveillance | <i>Note.</i> Clinical Utility and ASCO Biomarker testing Guidelines. ASCO biomarker testing recommendations are guided by the concept of clinical utility. A biomarker-based test is judged to have clinical utility if use of the test is demonstrated to result in a favorable balance of benefits to harms. This is addressed in the context of a randomized controlled trial or in the context of a prospective-retrospective study in which the biomarker is measured retrospectively on archived specimens after completion of a prospective clinical trial.                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                             |
|                                                                            | 2.1. ASCO does not recommend routine blood tests like CBC, chemistry panels, or tumor markers (CEA, CA 15-3, CA 27.29) for asymptomatic patients for the purpose of breast cancer surveillance. (Evidence quality: Low; Strength of recommendation: Conditional)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 91                                                                                                                                                                                                                                          |
|                                                                            | 2.2. ctDNA Recommendation (adapted from the ASCO ctDNA Testing Guideline): Fractional, percentage, or concentration-based measures of ctDNA or total cfDNA concentration are not recommended as a surrogate measure of disease burden to make treatment decisions outside the context of a clinical trial or clinical research. ctDNA testing cannot be recommended generally or, at this time, in any specific context for recurrence monitoring. (Evidence quality: Low; Strength of recommendation: Conditional)                                                                                                                                                                                                                                                                                    | 94                                                                                                                                                                                                                                          |
|                                                                            | <i>Qualifying Statement for Recommendation 2.2:</i> The identified evidence shows that ctDNA testing may have clinical validity in predicting the risk of a patient experiencing relapse, recurrence, or progression, and in detecting the presence of those events. However, no prospective studies were found that establish the clinical utility of ctDNA testing replacing or augmenting standard of care strategies for monitoring or surveillance. Without such evidence, the true benefit of ctDNA is highly uncertain, and there is the potential for harm in unfounded patient anxiety in response to false positive results.                                                                                                                                                                 |                                                                                                                                                                                                                                             |
| Supplemental surveillance imaging                                          | 3.1. Supplemental surveillance imaging with MRI may be offered and should be based on shared decision-making that considers individual patient characteristics (known germline mutation carriers, prior radiation therapy, additional factors in Recommendation 3.3), the relative benefits (reducing interval cancers), and harms (anxiety due to false positive findings, increased fear of recurrence, further investigations for incidental findings, time toxicity, and financial opportunity costs for patients and health systems) of more intensive surveillance, and patient preferences and values. There is no evidence to date for an overall survival benefit of MRI imaging in the post-treatment surveillance setting. (Evidence quality: Low; Strength of recommendation: Conditional) | 97                                                                                                                                                                                                                                          |
|                                                                            | <i>Qualifying Statement for Recommendation 3.1:</i> For patients at increased lifetime risk prior to primary breast cancer diagnosis (excluding those with germline mutations and history of radiation therapy with exposure to breast tissue), risk of subsequent breast cancer is mediated by patient, primary cancer, and treatment characteristics (refer to Recommendation 3.3 Additional Factors).                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                             |
|                                                                            | 3.2. Other supplemental imaging options include contrast enhanced mammography or breast ultrasound for patients with heterogeneously or extremely dense breasts and other risk factors, which may be offered after discussion of the relative benefits and harms. There is no evidence to date for an overall survival benefit of these imaging modalities in the post-treatment surveillance setting. (Evidence quality: Low; Strength of recommendation: Conditional)                                                                                                                                                                                                                                                                                                                                | 94                                                                                                                                                                                                                                          |
|                                                                            | 3.3. Supplemental Imaging Recommendations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 91                                                                                                                                                                                                                                          |
|                                                                            | Patient Characteristics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                             |
|                                                                            | Known germline mutation carriers and untested patients with first degree relatives with known germline mutations and intact breast tissue continue MRI <sup>a</sup> after primary breast cancer treatment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <i>BRCA1, BRCA2, PTEN, STK11, TP53</i> Consider: <i>ATM, BARD1, CDH1, CHEK2, NF1, PALB2, RAD51C, RAD51D</i>                                                                                                                                 |
|                                                                            | Radiation therapy with exposure to breast tissue, with intact breast tissue, continue MRI <sup>a</sup> after primary breast cancer treatment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Exposure between age 10 and 30                                                                                                                                                                                                              |
|                                                                            | Additional Factors: In the absence of above characteristics supporting supplemental MRI prior to breast cancer diagnosis, consider MRI <sup>a</sup> after completion of treatment for the following additional factors.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Age at diagnosis <50, invasive lobular histology, primary breast cancer diagnosis within 12 months of a negative screening mammogram, primary breast cancer not detected on screening mammogram, heterogeneously or extremely dense breasts |
|                                                                            | 3.4. Imaging surveillance for distant metastases is not recommended for the overall population of patients who have completed breast cancer treatment and who are asymptomatic. (Evidence quality: Low; Strength of recommendation: Conditional)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 97                                                                                                                                                                                                                                          |

NOTE. The strength of the recommendation is defined as follows, Strong: In recommendations for an intervention, the desirable effects of an intervention outweigh its undesirable effects. In recommendations against an intervention, the undesirable effects of an intervention outweigh its desirable effects. All or almost all informed people would make the recommended choice for or against an intervention; Conditional/Weak: In recommendations for an intervention, the desirable effects probably outweigh the undesirable effects, but appreciable uncertainty exists. In recommendations against an intervention, the undesirable effects probably outweigh the desirable effects, but appreciable uncertainty exists. Most informed people would choose the recommended course of action, but a substantial number would not.

Abbreviations: ACR, American College of Radiology; BCS, breast-conserving surgery; CA, cancer antigen; CDK4/6, cyclin-dependent kinase 4/6; CEA, carcinoembryonic antigen; ctDNA, circulating tumor DNA; ER, estrogen receptor; HER2, human epidermal growth factor receptor 2; HR, hormone receptor; MRI, magnetic resonance imaging; NA, not applicable; pCR, pathologic complete response; RCT, randomized controlled trial; TNBC, triple-negative breast cancer.

<sup>a</sup>Surveillance with contrast enhanced mammography if patient has contraindications to MRI, and ultrasound if contrast enhanced mammography is not available (Evidence quality: Low; Strength of recommendation: Conditional).



**FIG 1.** Post-treatment surveillance for breast cancer: low-intensity. APP, advanced practice provider; BC, breast cancer; DCIS, ductal carcinoma in situ; HER2, human epidermal growth factor receptor 2; HR, hormone receptor; TNBC, triple-negative breast cancer.

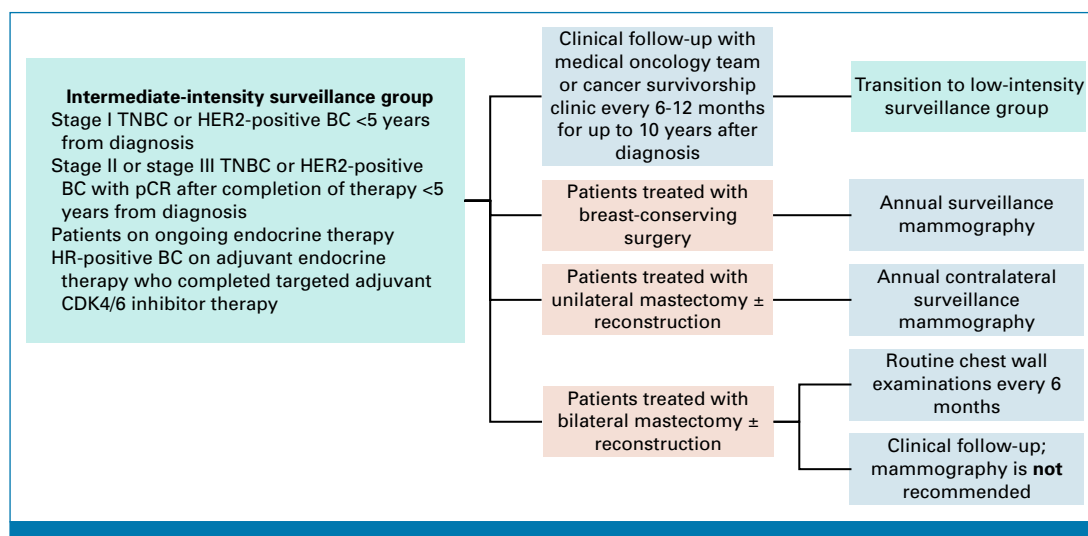
although its routine use is not yet established in clinical practice particularly in breast cancer.<sup>22</sup> This topic is further discussed in the subsequent section, Use of Blood-Based Biomarkers in Breast Cancer Post-treatment Surveillance.

### *Optimal Frequency and Duration of Monitoring for Recurrent Disease After Curative-Intent Primary Treatment*

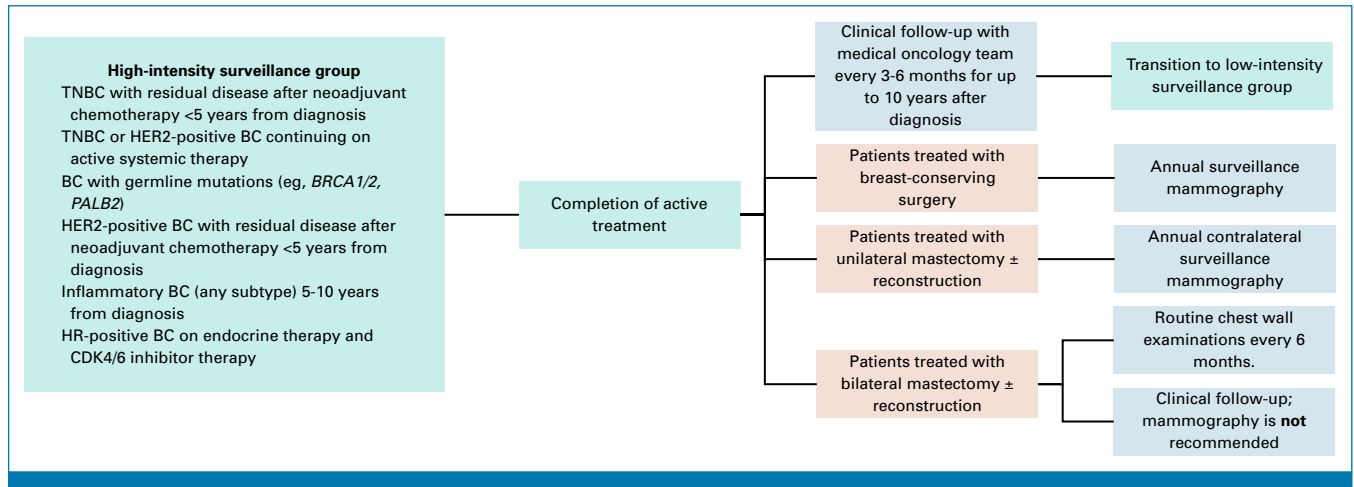
Optimal surveillance strategies and frequency may vary based on the patient's individual risk factors and treatment history. The frequency and duration of monitoring for recurrent disease in the adjuvant setting after curative-intent primary therapy vary based on the type of cancer and the guidelines followed.<sup>21</sup> There is a current paucity of evidence to support the effectiveness of intensive surveillance in terms of timeliness of recurrence detection or survival as opposed to less intensive surveillance.<sup>23-25</sup>

In Europe, similar opportunities were identified for improving the structured monitoring of patients with breast cancer after primary treatment. As an example, a nomogram from the Netherlands was developed as a prediction tool for

the risk of locoregional recurrences to personalize surveillance for breast cancer survivors and identify patients with a high risk of recurrence who might benefit from a less or more intensive follow-up after breast cancer.<sup>26</sup> The European Society for Medical Oncology (ESMO) has recommended regular follow-up visits every 3 months in the first 3 years post-treatment (every 6 months for low-risk early breast cancer), and every 6 months from years 4-5 and annually thereafter. However, ESMO noted that the interval of visits can be adapted to the risk of relapse and patient needs.<sup>23,27</sup> The Expert Panel recommends clinical examinations and patient history at regular intervals. However, the Panel recognized that there are limited data to support the routine clinical breast examination as a reliable mode of detection of early-stage breast cancer recurrence compared with imaging modalities, and limited value of clinical examination in diagnosing recurrence.<sup>28</sup> Therefore, surveillance programs may benefit from reduced focus on such a modality and a tailored approach based on intensity groups stratification is recommended (Table 4). Incorporating telehealth in follow-up care is also encouraged.



**FIG 2.** Post-treatment surveillance for breast cancer: intermediate-intensity. BC, breast cancer; CDK, cyclin-dependent kinase; HER2, human epidermal growth factor receptor 2; HR, hormone receptor; TNBC, triple-negative breast cancer.



**FIG 3.** Post-treatment surveillance for breast cancer: high-intensity. BC, breast cancer; CDK, cyclin-dependent kinase; HER2, human epidermal growth factor receptor 2; HR, hormone receptor; TNBC, triple-negative breast cancer.

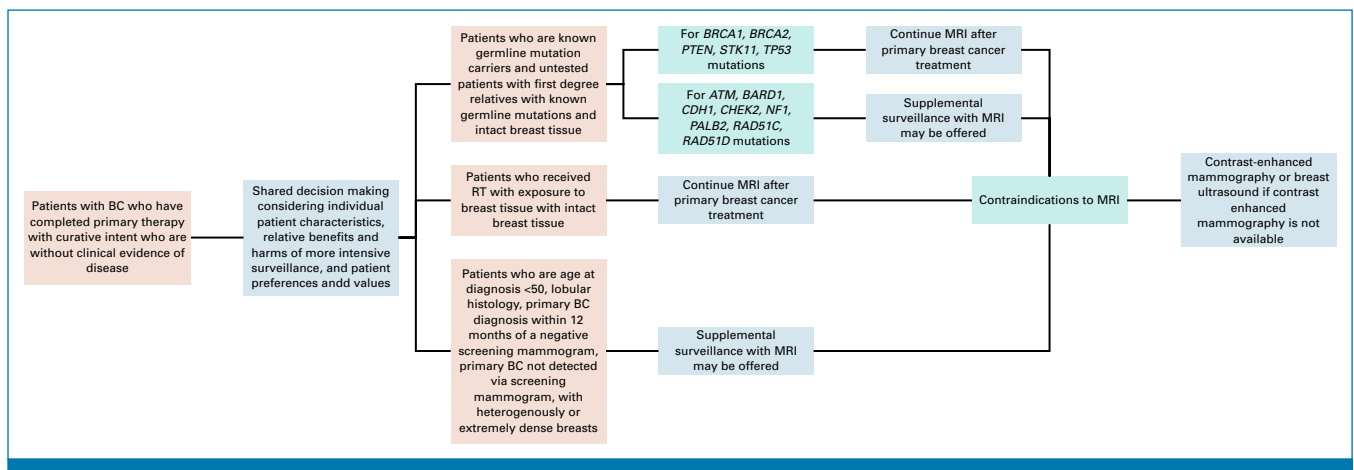
In the low-intensity group, annual clinical visits, in person or via virtual consultation, are recommended. In the intermediate-intensity surveillance group, clinical follow-up (including history and physical examination) with an oncology team or cancer survivorship clinic every 6–12 months is recommended for up to 10 years after diagnosis. And in the high-intensity group, after completing active treatment (eg, targeted therapy, immunotherapy, other than endocrine therapy), clinical follow-up with oncology team is recommended every 3–6 months for up to 10 years after diagnosis. Additional details on subsequent follow-up are included in [Table 4](#).

### Who Should Follow Patients in the Adjuvant Setting After Primary Treatment

The Expert Panel recognizes that clinical practices may vary regionally and globally. In Europe, there appear to be similar variations in the disciplines of involved health care professionals because of a lack of standard protocols. Opinions

on involving primary care physicians and general practice-based advanced practice providers (APPs) in aftercare and surveillance strongly differ as well. Less variation was observed for surveillance, which was guided by the Dutch national guideline.<sup>29</sup>

Although the decision on who should follow-up with patients is often based on the health care system's structure and the availability of specialists, a personalized approach based on the patient's specific needs and preferences is widely used.<sup>23,24,30</sup> Given the critical shortage in the oncology workforce relative to the number of patients who will need cancer care in the United States, the Expert Panel recognizes and appreciates the important role of other clinicians particularly APPs and survivorship clinics staffed with oncology APPs or nononcologists, such as primary care physicians in well-structured follow-up programs. It would be important to consider shared decision making and tailor the follow-up program to patient's care context and ensure real-time access to the patient's medical records.



**FIG 4.** Supplemental surveillance imaging. BC, breast cancer; MRI, magnetic resonance imaging; RT, radiation therapy.

## *Imaging for the Detection of Breast Cancer Recurrence After the Primary Treatment That Provides the Largest Benefit*

For patients who have undergone breast-conserving surgery (BCS), post-treatment surveillance mammography every 1–2 years is recommended for patients in the low-intensity surveillance group. This includes patients who are 50 years or older with low-risk features, including patients with small, low-grade, estrogen receptor (ER)–positive tumors, who underwent BCS, and who are recurrence free at 3 years postdiagnosis. Less frequent mammography is recommended based on expert consensus and on the findings of an RCT (Table 3).<sup>19</sup> The Expert Panel recommends annual surveillance mammography for patients with triple-negative breast cancer or human epidermal growth factor receptor 2 (HER2)–positive tumors. Of importance, decreasing disparities in receipt of timely surveillance mammography and follow-up care among patients with breast cancer is also of importance given the reported racial, ethnic disparities in the surveillance mammograms of breast cancer survivors.<sup>31</sup> Other related topics, including at what age to stop mammography screening and the use of supplemental imaging, are detailed in a subsequent section.

For other imaging recommendations, in an otherwise asymptomatic patient with no specific findings on clinical examination, routine imaging using intensive follow-up methods, such as radionuclide bone scans, chest radiographs, liver ultrasounds, computed tomography scans, and 18F-fluorodeoxyglucose positron emission tomography scans, are not recommended.<sup>21,31,32</sup> Routine imaging is not supported due to the potential for false positive results and their downstream consequences, unnecessary exposure to radiation, and potential for worsening financial hardship that extends well beyond active treatment.<sup>33,34</sup> Symptoms and signs that warrant further investigation in the adjuvant setting after curative intent primary therapy include any change in the size, shape, or appearance of the breast, nipple, or axilla. In addition, any new persistent pain or discomfort, no matter how long since initial breast cancer diagnosis, such as bone pain, abdominal pain, shortness of breath or chest pain, or persistent headaches or vision changes, should be promptly evaluated in the context of patient's history and clinical findings.

## **USE OF BLOOD-BASED BIOMARKERS IN BREAST CANCER POST-TREATMENT SURVEILLANCE**

This section addresses Recommendations 2.1 and 2.2.

### **Literature Review**

No systematic reviews, meta-analyses, or RCTs that met the inclusion criteria were identified by the systematic literature review. Consistent with previous ASCO guidelines,<sup>3,4</sup> the use of CBC, chemistry panels, CEA, CA 15–3, and CA 27.29 is not

recommended for routine surveillance of patients with breast cancer after completion of primary therapy.

In the ASCO ctDNA testing in solid tumors and lymphoma guideline,<sup>15</sup> three meta-analyses and nine additional trials were identified reporting on the association between ctDNA and recurrence. Of these, one meta-analysis, Gögenur et al<sup>35</sup> (10 studies, N = 727) focusing on patients who received neoadjuvant therapy included any patients with breast cancer (1 study, n = 26). Gögenur et al<sup>35</sup> reported a greater relative risk of distant recurrence for patients with detected ctDNA versus those without.

Of the nine additional trials, two were in early breast cancer (Liu and Liu<sup>36</sup> and Garcia-Murillas et al<sup>37</sup>). Liu and Liu<sup>36</sup> was a prospective cohort study analyzing urinary DNA as a predictor of relapse. The hazard ratio (HR) for disease-free survival in patients with higher urinary DNA concentrations at 9 months versus postsurgery was 1.51 (95% CI, 1.15 to 2.16). Garcia-Murillas et al<sup>37</sup> was a prospective cohort study assessing if plasma ctDNA can identify MRD after completion of primary treatment. In postsurgical samples, the HR for disease-free survival with ctDNA detected versus ctDNA not detected was 25.1 (95% CI, 4.08 to 130.5).

A supplemental targeted literature search conducted for the present guideline to identify RCTs or prospective-retrospective studies found no evidence to date of clinical utility to support the use of ctDNA testing in the breast cancer post-treatment surveillance setting.

### **Clinical Interpretation**

The detection of microscopic DNA tumor fragments or ctDNA has emerged as promising new biomarker in breast cancer as well as other malignancies.<sup>38</sup> Across multiple studies in all subtypes of early-stage breast cancer, the presence of tumor-informed ctDNA is associated with worse clinical outcomes and is clearly a negative prognostic indicator.<sup>39</sup> Detectable ctDNA at the time of diagnosis, after neoadjuvant therapy,<sup>40,41</sup> postsurgery or in surveillance follow-up has consistently identified patients at higher risk of relapse.<sup>42</sup> However, at present, there is no evidence for clinical utility of ctDNA in breast cancer. As such, the true benefit of ctDNA testing is highly uncertain, and there is potential for harm due to adverse events from therapies given unnecessarily or using and progressive through effective therapies before they were essential for use. Furthermore, although false positives are rare, they can increase fear of cancer recurrence and lead to unnecessary therapy. Before routine adoption of ctDNA in early-stage breast cancer, confirmation is needed that acting on ctDNA data by changing treatment ultimately improves patients' outcomes. Several trials in various settings of early-stage breast cancer are ongoing to generate data for addressing clinical utility questions. When ctDNA is used outside of clinical trials, patients are also at a risk of incurring unnecessary costs and anxiety when there is not clear evidence-based

approach for responding to these data (both false positives and negatives). Patients should be encouraged to participate in trials and studies that address the clinical utility of surveillance ctDNA monitoring along with shared decision making. The Panel acknowledges that this is a rapidly evolving area and these recommendations will be updated with emerging evidence in the future.

## SUPPLEMENTAL SURVEILLANCE IMAGING

This section addresses Recommendations 3.1 through 3.4.

### Literature Review

No systematic reviews, meta-analyses, or RCTs that met the inclusion criteria were identified by the systematic literature review. As noted earlier, mammographic surveillance recommendations in the low-intensity surveillance group were informed by findings of the Mammo-50 noninferiority phase III RCT.<sup>19</sup> A systematic review and meta-analysis identified by the search was excluded because of very high heterogeneity across the five mortality studies included. The recommendations offered represent the best clinical opinion of the Expert Panel members based on personal experience in the management of breast cancer and the input offered by members of the Consensus Panel.

### Clinical Interpretation

The Expert Panel adopted a risk-stratified approach to post-treatment surveillance imaging for subsequent breast cancer, including recommendations for mammographic surveillance recommendations as well as supplemental imaging. Post-treatment mammographic surveillance continues to be the mainstay imaging modality in clinical practice. Both standard digital mammography (DM) and digital breast tomosynthesis (DBT) may be used, as only around 48% of the 26,539 accredited mammography units in the United States are DBT. A comparison of DBT and DM for surveillance of asymptomatic women with personal history of breast cancer using a large sample drawn from US clinical practice with adjustment for population characteristics (80,077 mammograms from 28,042 women) found that while DBT had more favorable detection metrics compared with DM, as demonstrated by higher cancer detection rate (CDR), higher invasive CDR, and lower recall rate, the lack of significant reduction in interval cancer rate (3.0 and 3.1 per 1,000 for DBT and DM, respectively) indicates that both DBT and DM may be used for post-treatment mammographic surveillance.<sup>43</sup>

A remaining knowledge gap is when to stop mammography surveillance. The National Comprehensive Cancer Network Older Adult Oncology guideline<sup>44</sup> recommends that surveillance mammography should be guided by patient preference and life expectancy, with likely no benefit for those with a life expectancy of 5 years or less. A prospective study (ClinicalTrials.gov identifier: [NCT07123649](#)) is evaluating the effectiveness of a shared decision-making

toolkit to support older breast cancer survivors age 80 years and older in making informed decisions about continuing surveillance mammography. In contrast, due to a lack of currently available prospective data, the ACS and this Expert Panel continue to recommend surveillance mammography for all breast cancer survivors with residual breast tissue, regardless of age or life expectancy, despite limited evidence supporting its benefit in this population.<sup>44</sup> The value of mammography in older patients needs to be balanced against potential benefits, considering the low risk of local recurrence, the increased risk of overdiagnosis, false positives, and the absence of data to support mammographic screening in this age group.

Mammography alone has modest sensitivity for detecting subsequent breast cancers compared with screening mammography in the general population. Interval cancer rates in surveillance are fourfold higher than in screening,<sup>45</sup> motivating use of other imaging modalities for surveillance after breast cancer treatment, despite a paucity of evidence to support their effectiveness in improving long-term outcomes. A study of women with a personal history of breast cancer who received post-treatment surveillance imaging in US community practice, reported that approximately one-third underwent multiple breast imaging examinations per year during the first 2 years with mammography, magnetic resonance imaging (MRI), or both.<sup>46</sup> An analysis of surveillance imaging use among women with treated breast cancer indicated that those receiving MRIs were more likely to report White v non-White race, were more educated, and had higher income.<sup>47</sup> A more recent single institution study confirmed differential receipt of surveillance MRI by race, ethnicity, and insurance payer mix.<sup>48</sup> The observed differences in supplemental surveillance imaging use may reflect multiple undercharacterized drivers of health ranging from individual to health system levels, including social determinants of health, geography, referring clinician specialty, and local insurance coverage policies, which mediate both referral for imaging and attendance at appointments.

No randomized studies support supplemental surveillance imaging for asymptomatic patients with a history of breast cancer. Evidence for supplemental imaging surveillance effectiveness is limited; most studies are retrospective, heterogeneous in patient characteristics, influenced by local referral patterns, and with sample sizes much smaller than those of studies in the screening setting. Breast MRI studies in the surveillance setting demonstrate comparable diagnostic performance with larger studies in the screening setting, providing the strongest available evidence basis for supplemental MRI surveillance imaging.<sup>49</sup> A study of surveillance MRI (6,603 examinations in 2,809 women) reported an interval cancer rate of 1.5 per 1,000, stable over four surveillance rounds, lower than the interval cancer rate observed with surveillance mammography.<sup>50</sup> Beyond diagnostic accuracy, Yoon et al,<sup>51</sup> evaluated the association between surveillance MRI and advanced second breast cancers (size >2 cm, size >1 cm and triple negative or HER2 positive,

or lymph node positive) in a study of 1,062 patient pairs with propensity score matching. Receipt of surveillance MRI was associated with lower odds of advanced second breast cancer (odds ratio [OR], 0.41; 95% CI, 0.18 to 0.99), providing early evidence supporting MRI in improving post-treatment imaging outcomes. Other studies have sought to determine the optimal frequency of surveillance MRI. Park et al<sup>52</sup> found that in women with a personal history of breast cancer and negative initial MRI, 90% of second breast cancers were detected with MRI after a 2-year surveillance interval, and DeBruhl et al<sup>53</sup> found that no contralateral breast cancers were detected on MRI performed during a second year of MRI surveillance. While the sample sizes were relatively small, these studies indicate that MRI surveillance at 2-year intervals may be considered.

The high sensitivity of MRI in detecting breast cancer is based on its ability to detect cancer neoangiogenesis. More recently, contrast-enhanced mammography (CEM) has emerged as an alternative imaging modality which similarly uses an intravenously administered contrast agent to detect breast cancer at comparable sensitivity and specificity compared with MRI<sup>54</sup> and with additional potential advantages of shorter imaging acquisition time, greater access, and lower cost. While no studies to date have shown improved long-term outcomes in patients with a personal history of breast cancer, the BRAID RCT of supplemental screening<sup>55</sup> found CEM CDRs to be comparable with abbreviated breast MRI and superior to automated whole breast ultrasound in women with dense breasts.

Published studies of surveillance CEM have consistent results despite relatively small sample sizes. The Tomosynthesis or Contrast-Enhanced Mammography (TOCEM) study<sup>56</sup> of women with personal history of breast cancer in the United States found that adding CEM to DBT increased CDRs by 7.1 per 1,000 (95% CI, 3.2 to 13.4 per 1,000) to 22.8 per 1,000 for DBT plus CEM in the first year and by 3.8 per 1,000 (95% CI, 1.0 to 7.6 per 1,000) in incidence rounds. Study eligibility included women with both dense and nondense breasts. A study of incidence round surveillance CEM<sup>57</sup> in Australia reported a CDR of 15.4 per 1,000 and an interval cancer rate of 0.8 per 1,000. A small single-center study of patients with a personal history of breast cancer and dense breasts in Canada<sup>58</sup> reported a CDR of 18.6 per 1,000. The Panel recommends surveillance with CEM when MRI is clinically warranted, and patients have MRI contraindications or if MRI is otherwise unavailable.

Studies of surveillance with mammography, ultrasound, and MRI are few and tend to be retrospective analyses of single institution cohorts. In a relatively large retrospective study of 6,371 women within the first 5 years after breast cancer treatment,<sup>59</sup> MRI-based surveillance strategies had the highest CDR (10.2 per 1,000), greater than mammography or ultrasound alone (1.0 and 1.7 per 1,000, respectively) and comparable with mammography combined with ultrasound (8.7 per 1,000). 98.9% of women

underwent preoperative MRI before primary breast cancer treatment. In a small prospective multicenter study of multimodality imaging surveillance of women diagnosed with primary breast cancer before age 50 ( $n = 754$ ),<sup>60</sup> MRI had higher incremental cancer detection than ultrasound when added to mammography; ultrasound did not detect any breast cancers beyond those detected by mammography or MRI.

## LIMITATIONS OF THE RESEARCH AND FUTURE RESEARCH

As the debate on the optimal imaging surveillance modalities and follow-up care for breast cancer survivors continues, several key themes are clear. There is tremendous heterogeneity in breast cancer biology as well as in patient needs. For many years, clinicians have tried to adopt a one-size-fits-all model to survivorship care in breast cancer, and it is very clear that this approach is not optimal for meeting survivor needs nor is it efficient. The principle of accurately measuring risk at all points of care in the breast cancer care continuum is essential for personalized effective care delivery. Progress is slowly being made by challenging existing paradigms of what follow-up care should look like (digital and telehealth innovations), who should conduct the follow-up, biological predictors of relapse (ctDNA and other emerging biomarkers), and how to effectively optimize the overall health of breast cancer survivors whenever clinicians do evaluate them. At these crossroads of beginning to re-imagine the future of breast cancer care, the Expert Panel developed these recommendations to begin to tailor recommendations based on risk and patient preferences as an incremental effort.

Risk prediction models for patients with a personal history of breast cancer may increase the transparency, consistency, and evidence-based selection of patients for surveillance imaging and have a potential role in reducing observed disparities in use of supplemental surveillance imaging. Two emerging risk prediction models show promise for supporting risk-based surveillance imaging. The INFLUENCE 3.0 model,<sup>61</sup> which provides separate estimates for locoregional recurrence and contralateral breast cancer, is meant to guide decisions regarding surveillance mammography frequency. Model estimates of personalized risk are included in a decision aid that is under evaluation in the ongoing NABOR trial.<sup>62</sup> The Breast Cancer Surveillance Consortium (BCSC) Interval Second Breast Cancer model,<sup>63</sup> which provides 5-year risk of interval second breast cancer after a negative surveillance mammogram to guide selection of patients for supplemental surveillance imaging, demonstrates similar discriminatory accuracy to breast cancer risk prediction models currently in use for patients without a personal history of breast cancer. External validation of this model is an area of active investigation. To successfully advance risk-based surveillance imaging care, future studies should evaluate risk model implementation combined with additional approaches that improve access to care.

## PATIENT AND CLINICIAN COMMUNICATION

To optimize patient and clinician communication around breast cancer follow-up and surveillance after primary treatment, ongoing open communication and shared decision making is needed. The Discussion Points Between Patients and Clinicians box provides suggested points of discussion between physicians and patients for this purpose.

### DISCUSSION POINTS BETWEEN PATIENTS AND CLINICIANS

- Goals of follow-up and rationale for a risk-adapted approach
- Risks and benefits of the various options
- Comorbidities
- Use (or lack of use of) of blood-based markers and relevant evidence
- Common concerns include:
  - Recommended imaging strategies and associated worries
  - Psychosocial concerns and available resources
  - Financial and insurance considerations and resources
  - Survivorship planning, including long-term follow-up and monitoring

In this guideline, a novel risk-stratified approach to post-treatment surveillance for breast cancer is adopted to better meet the patients' needs, tailoring the intensity and frequency of postsurveillance imaging and follow-up visits. In some cases, transitioning to less intensive surveillance or to survivorship care might be uncomfortable for both clinicians and patients alike. Good patient-clinician communication might help with facilitating this shift as de-escalating breast cancer surveillance and transitioning to long-term survivorship care or escalating surveillance intensity, when appropriate, are aligned with and no different from tailoring surgery, radiotherapy, or chemotherapy to improve QoL and decrease toxicity, while maintaining high survival rates.

Optimal patient-clinician communication would be clear about the potential downside of the use of commercially available blood-based markers without robust relevant evidence to impact or help guide treatment decisions. This guideline attempts to facilitate some of these conversations for clinicians who are increasingly having to spend a significant amount of time explaining that the results of ctDNA tests ordered by other physicians that patients bring in, are not useful in guiding clinical care at this time. The Panel recognizes the patients' and clinicians' desire to find more options to detect early cancer recurrence, especially given that practices have changed in other cancer types such as in colon cancer where using ctDNA is now common in the adjuvant setting to guide treatment. However, more mature data from ongoing trials in breast cancer that are assessing ctDNA-guided adjuvant therapy in breast cancer are needed before safe and cost-effective implementation in clinical practice, particularly for patients at a high risk of recurrence

(DARE [NCT04567420](#), CATE [NCT06923527](#), LEADER [NCT03285412](#), CIPHER [NCT05333874](#), EXACTDNA003 [NCT06401421](#), SIGNAL-ER 101 [NCT07214532](#)).

This guideline also offers a risk-based approach to surveillance imaging and to supplemental modalities that offer increased cancer detection compared with mammography alone. As noted before, the Mammo-50 RCT indicated that there was no differences in any of the QoL scales between the two trial arms (mammogram at 1 year v 2 years) suggesting that less frequent mammographic surveillance, in the appropriate low-intensity surveillance group setting, does not adversely impact participants' QoL.<sup>20</sup> Careful discussion with patients about supplemental imaging is recommended as benefits must be balanced with possible harms of increased anxiety due to false-positive findings or the potential for further investigations of incidental findings.

For recommendations and strategies to optimize patient-clinician communication, see Patient-Clinician Communication: ASCO Guideline Update.<sup>64</sup> Antoni et al<sup>65</sup> have reviewed evidence for the effects of stress management interventions on psychological outcomes in cancer survivors. They note that anxiety in anticipation of post-treatment surveillance scans—the so-called scanxiety—and fear of cancer recurrence are common and are associated with significantly worse QoL outcomes.<sup>66-68</sup> Chronic stress in cancer survivors is also associated with dysregulated immune functioning, increased inflammation, and worsened physical health outcomes. Stress management interventions, cognitive-behavioral therapy in particular, are available that teach survivors skills to help them manage stress and allay the harmful effects of stress on their emotional well-being and are encouraged.<sup>65</sup>

### PATIENT ACCESS CONSIDERATIONS

In 2025, ASCO updated its guideline protocol template to include a patient access considerations section, allowing the Panel to consider patient access issues before starting the literature search.

Not all patients have consistent access to the resources needed to benefit from ASCO's expert recommendations on best practices for prevention, screening, palliative and supportive care, and disease management of cancer due to a multitude of challenges. These include lack of transportation, stable housing, food insecurity, health literacy, language barriers, distance from a dedicated cancer center, difficulty in finding child care, inability to miss work, high level of other comorbidities, insurance coverage, and cost of treatment and follow-up to name a few.<sup>69</sup> Additionally, some populations experience stigma along with barriers to cancer screening, prevention, and treatment that contribute to these cancer disparities.<sup>70</sup> Racial and ethnic disparities have been shown in the receipt of surveillance mammography in breast cancer survivors.<sup>31</sup>

Geographic disparities can also impact the quality-of-care patients receive.<sup>71</sup> Rural patients are more likely to have worse survivorship outcomes and experience higher mortality rates compared with nonrural patients. This can be attributed, in part, to a lower density of specialists and dedicated cancer centers, as only 21% of nonmetropolitan counties in the United States have one or more practicing oncologists.<sup>72</sup> The Expert Panel recognizes the important role of nononcologists, such as primary care and obstetrics/gynecology physicians and APPs, in follow-up programs that can broaden access to high-quality survivorship care.

Some effective interventions to improve survivorship care include patient navigation programs and telehealth. One study of survivorship care plans with nurse counseling reported significantly greater implementation of survivorship care.<sup>73</sup> One systematic review found that navigation increased mammography rates among breast cancer survivors.<sup>74</sup> Telehealth can potentially overcome geographic and access barriers, but may exacerbate disparities in patients with language barriers and limited access to technologies.

## MULTIPLE CHRONIC CONDITIONS

Patients with multiple chronic conditions (MCC), defined as two or more concurrent chronic illnesses, have additional complexities when clinicians are developing breast cancer surveillance and follow-up plans. Patients with MCC have historically been excluded from oncology clinical trials, and systematic measurement of comorbidities in cancer clinical trials is also lacking.<sup>75</sup> Taken together, these limitations restrict the evidence base for making informed decisions regarding breast cancer follow-up and surveillance in patients with MCC.<sup>76-78</sup> As a result, guidelines developed in more homogeneous populations are routinely applied to patients with MCC. A critical, and often underappreciated, consideration in this population is that comorbid conditions may impose a competing mortality risk that supersedes the risk from breast cancer itself. Most older breast cancer survivors with MCC will ultimately die from non-breast cancer-related causes, and specific guidelines for surveillance strategies in this population remains limited, with prospective data on the benefits and harms of continued surveillance mammography largely lacking.<sup>79-83</sup> When a patient's life expectancy is substantially limited by a comorbid condition, continued breast cancer surveillance may offer little meaningful benefit while contributing to patient burden, unnecessary procedures, and health care costs.<sup>80</sup> Decisions on mammographic surveillance should incorporate whether a patient's anticipated life expectancy is likely to be less than 5 years, regardless of their chronologic age.<sup>84</sup> Established methods for estimating life expectancy, such as geriatric assessment or ePrognosis,<sup>102</sup> can be used,<sup>85-87</sup> yet implementation of such individualized, life expectancy-informed approaches for patients with MCC remains a challenge. Ongoing clinical trials (eg, I-MAMMO [NCT07123649](#)) are seeking to address this gap.

## COST IMPLICATIONS

The financial toxicity of breast cancer treatment is well recognized as a significant and persistent burden for patients estimated to affect about 40% of survivors in high-income countries and more than 70% in low- and middle-income countries.<sup>33</sup> In one third or more of survivors of early breast cancer, financial hardship continues for many years beyond cessation of active treatment.<sup>34,88</sup> The lingering economic burden, effects on employability, and time away from work disproportionately affects younger women, and no doubt impacts compliance with surveillance, as does the overall increase in recent years in out of pocket costs of care.<sup>88-90</sup>

### Costs Related to Supplemental Surveillance Imaging

There is a paucity of data specifically addressing cost-benefit analyses for post-treatment surveillance imaging, including supplemental surveillance imaging. Underutilization of surveillance mammography after breast cancer treatment overall has been shown by declining use from 69.5% in year 1 to 61.0% by year 4 post-treatment.<sup>91</sup> Even in insured cohorts, 13% of patients did not undergo any breast imaging in the first year after surgery, rising to 19% by year 5.<sup>92</sup> Lack of imaging was independently associated with public or no insurance v private insurance ( $P < .005$ ), along with other sociodemographic factors including older age, Black race, and greater comorbidity.<sup>91</sup> Similarly, breast MRI rates during surveillance also decreased from 12.5% in year 1 to 5.8% by year 4.<sup>91</sup> Patients with Medicaid were more likely to undergo breast MRI than those with commercial insurance (OR, 1.57 [95% CI, 1.10 to 2.25]), suggesting that commercial insurance plans may have more restrictive coverage policies or higher cost-sharing for supplemental imaging.<sup>48</sup> This finding highlights potential disparities even among insured populations.

The evidence for specific policy interventions to improve access to supplemental surveillance imaging for breast cancer survivors is limited, with most research focusing on screening mammography rather than surveillance imaging. The Affordable Care Act's elimination of cost-sharing for screening mammography provides the strongest evidence for policy impact. Studies demonstrate that removing out-of-pocket costs increased screening mammography rates by approximately 6 percentage points among Medicare Advantage enrollees and reduced per-visit costs from \$25.43 to \$6.50 (74% reduction).<sup>93,94</sup> Among rural women, elimination of cost-sharing increased annual screening rates from 35% to 50% for women age 40-49 years and biennial screening from 56% to 66%.<sup>95</sup> These results suggest that policy interventions may increase adherence to evidence-based, guideline-recommended surveillance imaging care. However, data specifically addressing cost-benefit analyses for post-treatment surveillance are sparse. A recent observational cohort study of 812,164 women undergoing over two million screening episodes included 101,103 screening exams

in women with a personal history of breast cancer. In this subgroup, screening MRI led to substantially higher biopsy rates yet a lower CDR than mammography alone suggesting further investigation is needed to better select patients for whom post-treatment surveillance MRI is most likely have value.<sup>96</sup> Extrapolating from data on patient-clinician shared decision-making around costs of cancer treatment care, such discussions around the costs of post-treatment surveillance may help decrease out-of-pocket costs for patients while improving utilization of appropriate monitoring. Patients who had detailed cost discussions with clinicians did not seem to sacrifice the appropriate utilization of necessary cancer treatments.<sup>97</sup>

## GUIDELINE IMPLEMENTATION

ASCO guidelines are developed for implementation across health settings. Each ASCO guideline includes a community oncology member on the Panel. The roles of this member on the Panel are to assess the suitability of the recommendations to implementation in the community setting and to identify any other barriers to implementation of which a reader should be aware. Barriers to implementation include the need to increase awareness of the guideline recommendations among front line practitioners and survivors of cancer and caregivers, and also to provide adequate services in the face of limited resources.

For this guideline, clinicians and patients at a lower risk of recurrence who are accustomed to more frequent mammography and follow-up visits in oncology may be uncomfortable with less frequent mammography or with the idea of shifting care to a survivorship or primary care clinician. Good patient-clinician communication about the patient's risk and an understanding of the benefits to the patient of unnecessary follow-up (eg, avoidance of unnecessary anxiety about scans; benefits of being seen by a survivorship clinician who can also address long-term and late effects of treatment) is needed. Also, clinicians can consider whether they prefer to adopt a new care approach only with new patients for whom changing long-standing expectations and behaviors is not a concern.

The guideline recommendations table and accompanying tools (available on the [ASCO Publications website](https://ascopubs.org/doi/10.1200/JCO.24.00542)) were designed to facilitate implementation of recommendations. This guideline will be distributed widely. ASCO guidelines are most often published in the *Journal of Clinical Oncology*.

**ASCO believes that cancer clinical trials are vital to inform medical decisions and improve cancer care and that all patients should have the opportunity to participate.**

## RELATED ASCO GUIDELINES

- Palliative Care for Patients With Cancer<sup>98</sup> (<https://ascopubs.org/doi/10.1200/JCO.24.00542>)
- Patient-Clinician Communication<sup>64</sup> (<https://ascopubs.org/doi/10.1200/JCO-26-00118>)
- Practical Assessment and Management of Vulnerabilities in Older Patients Receiving Systemic Cancer Therapy<sup>86</sup> (<https://ascopubs.org/doi/10.1200/JCO.23.00933>)
- Management of Male Breast Cancer<sup>99</sup> (<https://ascopubs.org/doi/10.1200/JCO.19.03120>)
- Circulating Tumor DNA Testing in Solid Tumors and Lymphoma<sup>15</sup> (<https://ascopubs.org/doi/10.1200/OP-26-00311>)

## ADDITIONAL RESOURCES

For current information, including selected updates, supplements, and clinical tools and resources, visit the [ASCO Publications website](https://ascopubs.org/doi/10.1200/JCO.24.00542). The Data Supplement for this guideline includes literature search and review details. Guideline recommendations and algorithms are also available in the free ASCO Guidelines app (available for download in the [Apple App Store](https://appleappstore.com) and [Google Play Store](https://googleplaystore.com)). Listen to key recommendations and insights from panel members on the [ASCO Guidelines podcast](https://www.asco.org/practice-patients/guidelines/guideline-methodology). The Methodology Manual (available at <https://www.asco.org/practice-patients/guidelines/guideline-methodology>) provides additional information about the methods used to develop this guideline. Patient information is available at [www.cancer.org](https://www.cancer.org).

ASCO welcomes your comments on this guideline, including implementation challenges, new evidence, and how this guideline impacts you. To provide feedback, contact us at [guidelines@asco.org](mailto:guidelines@asco.org). Comments may be incorporated into a future guideline update. To submit new evidence or suggest a topic for guideline development, complete the form available at <https://www.asco.org/practice-patients/guidelines>.

## INCLUSIVE LANGUAGE

ASCO is committed to promoting the health and well-being of all patients. ASCO guidelines are intended to apply to, and be discussed clearly and compassionately with, all patients. For this reason, guideline authors use appropriately inclusive language. In instances in which the guideline draws upon data based on research in a specified population (eg, studies regarding women with breast cancer), the guideline authors describe the characteristics and results of the research as reported.

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Z.N. and M.L. were Expert Panel Co-Chairs.

## EDITOR'S NOTE

This American Society of Clinical Oncology (ASCO) Clinical Practice Guideline Update provides recommendations, with comprehensive

## REFERENCES

1. Recommended breast cancer surveillance guidelines. American Society of Clinical Oncology. *J Clin Oncol* 15:2149-2156, 1997
2. Smith TJ, Davidson NE, Schapira DV, et al: American Society of Clinical Oncology 1998 update of recommended breast cancer surveillance guidelines. *J Clin Oncol* 17:1080-1082, 1999
3. Khatcheressian JL, Wolff AC, Smith TJ, et al: American Society of Clinical Oncology 2006 update of the breast cancer follow-up and management guidelines in the adjuvant setting. *J Clin Oncol* 24:5091-5097, 2006
4. Khatcheressian JL, Hurley P, Bantug E, et al: Breast cancer follow-up and management after primary treatment: American Society of Clinical Oncology clinical practice guideline update. *J Clin Oncol* 31:961-965, 2013
5. Mayer DK, Alfano CM: Personalized risk-stratified cancer follow-up care: Its potential for healthier survivors, happier clinicians, and lower costs. *J Natl Cancer Inst* 111:442-448, 2019
6. Alfano CM, Mayer DK, Bhatia S, et al: Implementing personalized pathways for cancer follow-up care in the United States: Proceedings from an American Cancer Society-American Society of Clinical Oncology summit. *CA Cancer J Clin* 69:234-247, 2019
7. Erikson C, Salsberg E, Forte G, et al: Future supply and demand for oncologists: Challenges to assuring access to oncology services. *J Oncol Pract* 3:79-86, 2007
8. Yang W, Williams JH, Hogan PF, et al: Projected supply of and demand for oncologists and radiation oncologists through 2025: An aging, better-insured population will result in shortage. *J Oncol Pract* 10:39-45, 2014
9. Kirkwood MK, Balogh EP, Accordini MK, et al: Where have we been and where are we going? The state of the hematology and medical oncologist workforce in America. *JCO Oncol Pract* 21:1775-1785, 2025
10. National Center for Health Workforce Analysis, 2025, <https://bhw.hrsa.gov/data-research/review-health-workforce-research>
11. Alfano CM, Jefford M, Maher J, et al: Building personalized cancer Follow-up care pathways in the United States: Lessons learned from implementation in England, Northern Ireland, and Australia. *Am Soc Clin Oncol Educ Book* 39:625-639, 2019
12. Macmillan Cancer Support: Evaluation of the Transforming Cancer Follow-up Programme in Northern Ireland: Final Report United Kingdom, PwC, 2015
13. National Health Service England: Implementing personalised stratified follow up pathways, 2020. <https://www.england.nhs.uk/publication/implementing-personalised-stratified-follow-up-pathways/>
14. Jefford M, Howell D, Li Q, et al: Improved models of care for cancer survivors. *Lancet* 399:1551-1560, 2022
15. Lockwood CM, Messersmith HJ, Kim AS, et al: Circulating tumor DNA testing in solid tumors and lymphoma: ASCO guideline. *JCO Oncol Pract* 10.1200/OP-26-00311
16. Loblaw DA, Prestrud AA, Somerfield MR, et al: American Society of Clinical Oncology Clinical practice guidelines: Formal systematic review-based consensus methodology. *J Clin Oncol* 30:3136-3140, 2012
17. Higgins JPT: Cochrane Collaboration: Cochrane Handbook for Systematic Reviews of Interventions (ed 2). Hoboken, NJ, Wiley-Blackwell, 2019
18. Balshem H, Helfand M, Schunemann HJ, et al: GRADE guidelines: 3. Rating the quality of evidence. *J Clin Epidemiol* 64:401-406, 2011
19. Dunn JA, Donnelly P, Elbeltagi N, et al: Annual versus less frequent mammographic surveillance in people with breast cancer aged 50 years and older in the UK (Mammo-50): A multicentre, randomised, phase 3, non-inferiority trial. *Lancet* 405:396-407, 2025
20. Marshall A, Donnelly P, Elbeltagi N, et al: Quality of life and patient reported outcomes in the UK Mammo-50 randomised trial of annual versus less frequent mammographic surveillance in people with breast cancer aged 50 years and over. *Health Qual Life Outcomes* 23:67, 2025
21. Kyriazoglou A, Zagouri F, Fotiou D, et al: Discrepancies of current recommendations in breast cancer follow-up: A systematic review. *Breast Cancer* 26:681-686, 2019
22. Mittal A, Molto Valiente C, Tamimi F, et al: Utility of ctDNA in predicting relapse in solid tumors after curative therapy: A meta-analysis. *JNCI Cancer Spectr* 7:pkad040, 2023
23. Moschetti I, Ciniqini M, Lamberti M, et al: Follow-up strategies for women treated for early breast cancer. *Cochrane Database Syst Rev* 2016, 2016. CD001768
24. Hoeg BL, Bidstrup PE, Karlsen RV, et al: Follow-up strategies following completion of primary cancer treatment in adult cancer survivors. *Cochrane Database Syst Rev* 2019:CD012425, 2019
25. Lafranconi A, Pylkanen L, Deandrea S, et al: Intensive follow-up for women with breast cancer: Review of clinical, economic and patient's preference domains through evidence to decision framework. *Health Qual Life Outcomes* 15:206, 2017
26. Witteveen A, Vliegen IM, Sonke GS, et al: Personalisation of breast cancer follow-up: A time-dependent prognostic nomogram for the estimation of annual risk of locoregional recurrence in early breast cancer patients. *Breast Cancer Res Treat* 152:627-636, 2015

review and analyses of the relevant literature for each recommendation. Additional information, including a supplement with additional evidence tables, clinical tools, and resources can be found on the [ASCO Publications website](https://www.asco.org/publications). Patient information is available at [www.cancer.org](http://www.cancer.org).

## AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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27. Loibl S, Andre F, Bachelot T, et al: Early breast cancer: ESMO clinical practice guideline for diagnosis, treatment and follow-up. *Ann Oncol* 35:159-182, 2024
28. Horan J, Reid C, Boland MR, et al: Assessing mode of recurrence in breast cancer to identify an optimised follow-up pathway: 10-year institutional review. *Ann Surg Oncol* 30:6117-6124, 2023
29. Ankersmid JW, van Hoeve JC, Strobbe LJA, et al: Follow-up after breast cancer: Variations, best practices, and opportunities for improvement according to health care professionals. *Eur J Cancer Care (Engl)* 30:e13505, 2021
30. Ankersmid JW, Siesling S, Strobbe LJA, et al: Supporting shared decision-making about surveillance after breast cancer with personalized recurrence risk calculations: Development of a patient decision aid using the international patient decision AIDS standards development process in combination with a mixed methods design. *JMIR Cancer* 8:e38088, 2022
31. Advani P, Advani S, Nayak P, et al: Racial/ethnic disparities in use of surveillance mammogram among breast cancer survivors: A systematic review. *J Cancer Surviv* 16:514-530, 2022
32. Galjart B, Hoppener DJ, Aerts J, et al: Follow-up strategy and survival for five common cancers: A meta-analysis. *Eur J Cancer* 174:185-199, 2022
33. Yin J, Wang C, Hou Y, et al: Financial toxicity among patients with breast cancer: A systematic review and meta-analysis. *Oncologist* 31:oyaf417, 2026
34. Myers SP, Zheng Y, Dibble K, et al: Financial difficulty over time in young adults with breast cancer. *JAMA Netw Open* 7:e2446091, 2024
35. Gögenur M, Hadi NA, Qvortrup C, et al: ctDNA for risk of recurrence assessment in patients treated with neoadjuvant treatment: A systematic review and meta-analysis. *Ann Surg Oncol* 29:8666-8674, 2022
36. Liu Z, Liu W.: Association of urinary and plasma DNA in early breast cancer patients and its links to disease relapse. *Clin Transl Oncol* 20:1053-1060, 2018
37. Garcia-Murillas I, Schiavon G, Weigelt B, et al: Mutation tracking in circulating tumor DNA predicts relapse in early breast cancer. *Sci Transl Med* 7:302ra133, 2015
38. Schlamm I, Tolane SM, Lin NU, et al: Circulating tumor DNA in early breast cancer: A review. *JAMA Oncol* 12:773-784, 2026
39. Nader-Marta G, Monteforte M, Agostinetto E, et al: Circulating tumor DNA for predicting recurrence in patients with operable breast cancer: A systematic review and meta-analysis. *ESMO Open* 9:102390, 2024
40. Elliott MJ, Howarth K, Main S, et al: Ultrasensitive detection and monitoring of circulating tumor DNA using structural variants in early-stage breast cancer. *Clin Cancer Res* 31:1520-1532, 2025
41. Parsons HA, Blewett T, Chu X, et al: Circulating tumor DNA association with residual cancer burden after neoadjuvant chemotherapy in triple-negative breast cancer in TBCRC 030. *Ann Oncol* 34:899-906, 2023
42. Shaw JA, Page K, Wren E, et al: Serial postoperative circulating tumor DNA assessment has strong prognostic value during long-term Follow-Up in patients with breast cancer. *JCO Precis Oncol* 8:e2300456, 2024
43. Lee JM, Ichikawa L, Su YR, et al: Comparative accuracy of digital breast tomosynthesis and digital mammography for surveillance after breast cancer treatment. *J Natl Compr Canc Netw* 23:445-452, 2025
44. Freedman RA, Minami CA, Winer EP, et al: Individualizing surveillance mammography for older patients after treatment for early-stage breast cancer: Multidisciplinary expert panel and International Society of Geriatric Oncology consensus statement. *JAMA Oncol* 7:609-615, 2021
45. Lee CI, Abraham L, Miglioretti DL, et al: National performance benchmarks for screening digital breast tomosynthesis: Update from the Breast Cancer Surveillance Consortium. *Radiology* 307:e222499, 2023
46. Henderson LM, Ichikawa L, Buist DSM, et al: Patterns of breast imaging use among women with a personal history of breast cancer. *J Gen Intern Med* 34:2098-2106, 2019
47. Wernli KJ, Ichikawa L, Kerlikowske K, et al: Surveillance breast MRI and mammography: Comparison in women with a personal history of breast cancer. *Radiology* 292:311-318, 2019
48. Kakani P, Ro V, McGuinness JE, et al: Factors associated with receipt of surveillance breast MRI among racially and ethnically diverse women with operable breast cancer. *Breast Cancer Res Treat* 215:27, 2025
49. Haas CB, Nekhlyudov L, Lee JM, et al: Surveillance for second breast cancer events in women with a personal history of breast cancer using breast MRI: A systematic review and meta-analysis. *Breast Cancer Res Treat* 181:255-268, 2020
50. Kim GR, Cho N, Kim SY, et al: Interval cancers after negative supplemental screening breast MRI results in women with a personal history of breast cancer. *Radiology* 300:314-323, 2021
51. Yoon J, Han K, Nahm S, et al: Surveillance breast MRI in women with a history of breast cancer: Association with occurrence of advanced second breast cancer. *Radiology* 314:e240119, 2025
52. Park VV, Kim MJ, Kim GR, et al: Outcomes following negative screening MRI results in Korean women with a personal history of breast cancer: Implications for the next MRI interval. *Radiology* 300:303-311, 2021
53. Debruhi ND, Lee SJ, Mahoney MC, et al: MRI evaluation of the contralateral breast in women with recently diagnosed breast cancer: 2-year follow-up. *J Breast Imaging* 2:50-55, 2020
54. Neeter L, Robbe MMQ, van Nijntjen TJA, et al: Comparing the diagnostic performance of contrast-enhanced mammography and breast MRI: A systematic review and meta-analysis. *J Cancer* 14:174-182, 2023
55. Gilbert FJ, Payne NR, Allajbeu I, et al: Comparison of supplemental breast cancer imaging techniques-interim results from the BRAID randomised controlled trial. *Lancet* 405:1935-1944, 2025
56. Berg WA, Berg JM, Bandos AI, et al: Addition of contrast-enhanced mammography to tomosynthesis for breast cancer detection in women with a personal history of breast cancer: Prospective TOCEM trial interim analysis. *Radiology* 311:e231991, 2024
57. Matheson J, Elder K, Nickson C, et al: Contrast-enhanced mammography for surveillance in women with a personal history of breast cancer. *Breast Cancer Res Treat* 208:293-305, 2024
58. Gosein M, Khoo E, Yong-Hing C, et al: Surveillance contrast-enhanced mammography in patients with dense breasts and a personal history of breast cancer. *Clin Breast Cancer* 26:17-26, 2026
59. Kim MK, Park MS, Go MG, et al: Surveillance outcomes by imaging methods in the first 5 years after breast cancer surgery. *Korean J Radiol* 26:532-545, 2025
60. Cho N, Han W, Han BK, et al: Breast cancer screening with mammography plus ultrasonography or magnetic resonance imaging in women 50 years or younger at diagnosis and treated with breast conservation therapy. *JAMA Oncol* 3:1495-1502, 2017
61. Van Maaren MC, Huetting TA, van Uden DJP, et al: The INFLUENCE 3.0 model: Updated predictions of locoregional recurrence and contralateral breast cancer, now also suitable for patients treated with neoadjuvant systemic therapy. *Breast* 79:103829, 2025
62. Klaassen-Dekker A, Drossaert CHC, Van Maaren MC, et al: Personalized surveillance and aftercare for non-metastasized breast cancer: The NABOR study protocol of a multiple interrupted time series design. *BMC Cancer* 23:1112, 2023
63. Hubbard RA, Su YR, Bowles EJA, et al: Predicting five-year interval second breast cancer risk in women with prior breast cancer. *J Natl Cancer Inst* 116:929-937, 2024
64. Gilligan T, Bohlke K, Alpert AB, et al: Patient-clinician communication: ASCO guideline update. *J Clin Oncol* 44:1040-1057, 2026
65. Antoni MH, Moreno PI, Penedo FJ.: Stress management interventions to facilitate psychological and physiological adaptation and optimal health outcomes in cancer patients and survivors. *Annu Rev Psychol* 74:423-455, 2023
66. Bui KT, Liang R, Kiely BE, et al: Scanxiety: A scoping review about scan-associated anxiety. *BMJ Open* 11:e043215, 2021
67. Koch L, Jansen L, Brenner H, et al: Fear of recurrence and disease progression in long-term ( $\geq 5$  years) cancer survivors—A systematic review of quantitative studies. *Psychooncology* 22:1-11, 2013
68. Simard S, Thewes B, Humphris G, et al: Fear of cancer recurrence in adult cancer survivors: A systematic review of quantitative studies. *J Cancer Surviv* 7:300-322, 2013
69. Hendren S, Chin N, Fisher S, et al: Patients' barriers to receipt of cancer care, and factors associated with needing more assistance from a patient navigator. *J Natl Med Assoc* 103:701-710, 2011
70. Alpert AB, Scout NFN, Schabath MB, et al: Gender- and sexual orientation- based inequities: Promoting inclusion, visibility, and data accuracy in oncology. *Am Soc Clin Oncol Educ Book* 42:1-17, 2022
71. Sogade OO, Margenthaler JA: Evaluating factors associated with advanced-stage breast cancer presentation in rural patients in the National Cancer Database. *J Am Coll Surg* 242:1138-1146, 2026
72. Levit LA, Byatt L, Lyss AP, et al: Closing the rural cancer care gap: Three institutional approaches. *JCO Oncol Pract* 16:422-430, 2020
73. Maly RC, Liang LJ, Liu Y, et al: Randomized controlled trial of survivorship care plans among low-income, predominantly Latina breast cancer survivors. *J Clin Oncol* 35:1814-1821, 2017
74. Baik SH, Gallo LC, Wells KJ: Patient navigation in breast cancer treatment and survivorship: A systematic review. *J Clin Oncol* 34:3686-3696, 2016
75. Unger JM, Hershman DL, Fleury ME, et al: Association of patient comorbid conditions with cancer clinical trial participation. *JAMA Oncol* 5:326-333, 2019
76. Lichtman SM, Harvey RD, Damiette Smit MA, et al: Modernizing clinical trial eligibility criteria: Recommendations of the American Society of Clinical Oncology-Friends of Cancer research organ dysfunction, prior or concurrent malignancy, and Comorbidities Working Group. *J Clin Oncol* 35:3753-3759, 2017
77. Williams GR, Mackenzie A, Magnuson A, et al: Comorbidity in older adults with cancer. *J Geriatr Oncol* 7:249-257, 2016
78. Williams GR, Deal AM, Lund JL, et al: Patient-reported comorbidity and survival in older adults with cancer. *Oncologist* 23:433-439, 2018
79. Li H, Gondara L, Speers C, et al: Causes of death in older patients with high-risk breast cancer: What are the most significant contributors? *J Geriatr Oncol* 16:102348, 2025
80. Freedman RA, Keating NL, Partridge AH, et al: Surveillance mammography in older patients with breast cancer-can we ever stop?: A review. *JAMA Oncol* 3:402-409, 2017
81. Satariano WA, Ragland DR.: The effect of comorbidity on 3-year survival of women with primary breast cancer. *Ann Intern Med* 120:104-110, 1994
82. Yancik R, Wesley MN, Ries LA, et al: Effect of age and comorbidity in postmenopausal breast cancer patients aged 55 years and older. *JAMA* 285:885-892, 2001
83. Schonberg MA, Marcantonio ER, Li D, et al: Breast cancer among the oldest old: Tumor characteristics, treatment choices, and survival. *J Clin Oncol* 28:2038-2045, 2010
84. VanderWalde N, Jaggi R, Dotan E, et al: NCCN guidelines insights: Older adult oncology, version 2.2016. *J Natl Compr Canc Netw* 14:1357-1370, 2016
85. Magnuson A, Loh KP, Stauffer F, et al: Geriatric assessment for the practicing clinician: The why, what, and how. *CA Cancer J Clin* 74:496-518, 2024

86. Dale W, Klepin HD, Williams GR, et al: Practical assessment and management of vulnerabilities in older patients receiving systemic cancer therapy: ASCO guideline update. *J Clin Oncol* 41: 4293-4312, 2023
87. Kimmick GG, Major B, Clapp J, et al: Using ePrognosis to estimate 2-year all-cause mortality in older women with breast cancer: Cancer and Leukemia Group B (CALGB) 49907 and 369901 (Alliance A151503). *Breast Cancer Res Treat* 163:391-398, 2017
88. Jagi R, Pottow JA, Griffith KA, et al: Long-term financial burden of breast cancer: Experiences of a diverse cohort of survivors identified through population-based registries. *J Clin Oncol* 32: 1269-1276, 2014
89. Politi MC, Yen RW, Elwyn G, et al: Women who are young, non-White, and with lower socioeconomic status report higher financial toxicity up to 1 year after breast cancer surgery: A mixed-effects regression analysis. *Oncologist* 26:e142-e152, 2021
90. Schnipper LE, Davidson NE, Wollins DS, et al: American Society of Clinical Oncology statement: A conceptual framework to assess the value of cancer treatment options. *J Clin Oncol* 33: 2563-2577, 2015
91. Adesoye T, Schumacher JR, Neuman HB, et al: Use of breast imaging after treatment for locoregional breast cancer (AFT-01). *Ann Surg Oncol* 25:1502-1511, 2018
92. Ruddy KJ, Sangaralingham L, Freedman RA, et al: Adherence to guidelines for breast surveillance in breast cancer survivors. *J Natl Compr Canc Netw* 16:526-534, 2018
93. Trivedi AN, Leyva B, Lee Y, et al: Elimination of cost sharing for screening mammography in Medicare advantage plans. *N Engl J Med* 378:262-269, 2018
94. Kirby JB, Davidoff AJ, Basu J.: The ACA's zero cost-sharing mandate and trends in out-of-pocket expenditures on well-child and screening mammography visits. *Med Care* 54:1056-1062, 2016
95. Peppercorn J, Horick N, Houck K, et al: Impact of the elimination of cost sharing for mammographic breast cancer screening among rural US women: A natural experiment. *Cancer* 123: 2506-2515, 2017
96. Buist DSM, Abraham L, Lee CI, et al: Breast biopsy intensity and findings following breast cancer screening in women with and without a personal history of breast cancer. *JAMA Intern Med* 178: 458-468, 2018
97. Hong YR, Salloum RG, Yadav S, et al: Patient-provider discussion about cancer treatment costs and out-of-pocket spending: Implications for shared decision making in cancer care. *Value Health* 23:1592-1598, 2020
98. Sanders JJ, Temin S, Ghoshal A, et al: Palliative care for patients with cancer: ASCO guideline update. *J Clin Oncol* 42:2336-2357, 2024
99. Hassett MJ, Somerfield MR, Baker ER, et al: Management of Male breast cancer: ASCO guideline. *J Clin Oncol* 38:1849-1863, 2020
100. Schunemann HJ, Wiercioch W, Etzeandía I, et al: Guidelines 2.0: Systematic development of a comprehensive checklist for a successful guideline enterprise. *CMAJ* 186:E123-E142, 2014
101. Expert Panel on Breast Imaging, Brown A, Lourenco AP, et al: ACR Appropriateness Criteria transgender breast cancer screening. *J Am Coll Radiol* 18:S502-S515, 2021
102. University of California San Francisco: ePrognosis. <https://eprognosis.ucsf.edu/about.php>

## AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

### Breast Cancer Follow-Up and Surveillance After Primary Treatment: ASCO Guideline Update

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Open Payments is a public database containing information reported by companies about payments made to US-licensed physicians ([Open Payments](https://openpaymentsdata.cms.gov/physician/)).

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## APPENDIX 2. GUIDELINE AND CONFLICTS OF INTEREST

The Expert Panel was assembled in accordance with ASCO's Conflict of Interest Policy Implementation for Clinical Practice Guidelines ("Policy," found at <https://www.asco.org/practice-patients/guidelines/guideline-methodology>). All members of the Expert Panel completed ASCO's disclosure form, which requires disclosure of financial and other interests, including relationships with commercial entities that are reasonably likely to experience direct regulatory or commercial impact as a result of promulgation of the guideline. Categories for disclosure include employment; leadership; stock or other ownership; honoraria, consulting or advisory role; speaker's bureau; research funding; patents, royalties, other intellectual property; expert testimony; travel, accommodations, expenses; and other relationships. In accordance with the Policy, the majority of the members of the Expert Panel did not disclose any relationships constituting a conflict under the Policy.

**TABLE A1. Breast Cancer Follow-Up and Surveillance After Primary Treatment Guideline Expert Panel Membership**

| Name                                | Affiliation                                                                                | Role or Area of Expertise                               |
|-------------------------------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------|
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| Zeina Nahleh, MD (Co-Chair)         | Cleveland Clinic Florida, Weston Hospital, Weston, FL                                      | Medical oncology                                        |
| Helen K. Chew, MD                   | UC Davis Comprehensive Cancer Center, Sacramento, CA                                       | Medical oncology                                        |
| Stephen K. L. Chia, MD              | Division of Medical Oncology, Vancouver Cancer Center, Vancouver, BC, Canada               | Medical oncology                                        |
| Tina J. Hieken, MD                  | Mayo Clinic, Rochester, MN                                                                 | Surgical oncology                                       |
| Ellen Landsberger, MD               | New York City, NY                                                                          | Patient advocacy                                        |
| Janie M. Lee, MD, MSc               | University of Washington School of Medicine and Fred Hutchinson Cancer Center, Seattle, WA | Radiology                                               |
| Bora Lim, MD                        | MD Anderson Cancer Center, Houston, TX                                                     | Medical oncology                                        |
| Allison Magnuson, DO, MS            | University of Rochester Medical Center, Rochester, NY                                      | Geriatric oncology                                      |
| Sitna Ali Mwanzi, MBChB, MSc        | Kenyatta National Hospital, Nairobi, Kenya                                                 | Medical oncology (LMIC Member)                          |
| Heather A. Parsons, MD, MPH         | Fred Hutchinson Cancer Center, Seattle, WA                                                 | Medical oncology                                        |
| Debra Patt, MD, PhD, MBA            | Texas Oncology, Austin, TX                                                                 | Medical oncology (community oncology)                   |
| Ruta D. Rao, MD                     | Rush University Cancer Center, Chicago, IL                                                 | Medical oncology                                        |
| Jasmine S. Sukumar, MD              | MD Anderson Cancer Center, Houston, TX                                                     | Medical oncology                                        |
| Hyuna Sung, PhD                     | American Cancer Society, Atlanta, GA                                                       | Clinical epidemiology (ACS representative)              |
| Gustavo Werutsky, MD                | Latin American Cooperative Oncology Group, Porto Alegre, Brazil                            | Medical oncology                                        |
| Brittany E. Harvey, BS              | American Society of Clinical Oncology (ASCO), Alexandria, VA                               | ASCO Practice Guideline Staff (Health Research Methods) |
| Mark R. Somerfield, PhD             | American Society of Clinical Oncology (ASCO), Alexandria, VA                               | ASCO Practice Guideline Staff (Health Research Methods) |

**TABLE A2. Guideline Consensus Panel Membership**

| Name                                      | Affiliation                                                                                          |
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| Tina J. Hieken, MD <sup>a</sup>           | Mayo Clinic, Rochester, MN                                                                           |
| Regina A. Jacob, MD, MSc                  | Northwell Cancer Institute, New Hyde Park, NY                                                        |
| James Khatcheressian, MD                  | Virginia Cancer Institute, Richmond, VA                                                              |
| Bogda Koczwara, BM BS                     | Australian Research Centre for Cancer Survivorship, University of New South Wales, Sydney, Australia |
| Ellen Landsberger, MD <sup>a</sup>        | New York City, NY                                                                                    |
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**NOTE.** Disclosures of potential conflicts of interest provided by the consensus panel members are available in the guideline supplement.

<sup>a</sup>Member of ASCO Expert Panel.

**TABLE A3. Recommendation Rating Definitions**

| Term                       | Definition                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Quality of evidence        |                                                                                                                                                                                                                                                                                                                                                                                                             |
| High                       | We are very confident that the true effect lies close to that of the estimate of the effect.                                                                                                                                                                                                                                                                                                                |
| Moderate                   | We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.                                                                                                                                                                                                                     |
| Low                        | Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect.                                                                                                                                                                                                                                                                           |
| Very low                   | We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect.                                                                                                                                                                                                                                                                 |
| Strength of recommendation |                                                                                                                                                                                                                                                                                                                                                                                                             |
| Strong                     | In recommendations for an intervention, the desirable effects of an intervention outweigh its undesirable effects.<br>In recommendations against an intervention, the undesirable effects of an intervention outweigh its desirable effects.<br>All or almost all informed people would make the recommended choice for or against an intervention.                                                         |
| Conditional/weak           | In recommendations for an intervention, the desirable effects probably outweigh the undesirable effects, but appreciable uncertainty exists.<br>In recommendations against an intervention, the undesirable effects probably outweigh the desirable effects, but appreciable uncertainty exists.<br>Most informed people would choose the recommended course of action, but a substantial number would not. |

NOTE. GRADE Handbook<sup>100</sup>